



Gorkha North Field Report



CRS Shelter and Settlements Team

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Acronyms

VDC - Village development committees

CRS - Catholic Relief Services

RCC - Reinforced Cast Concrete

GoN - Government of Nepal

PDRF - Post Disaster Recovery FRamework

PDNA - Post Disaster Needs Assessment

NRA - NAtional Reconstruction Authority

DUDBC - Department of Urban Development + Building Control

CGI - Corrugated Galvanised Iron

MCAP - Manasulu Conservation Area Project

PIN - People In Need

HRRP - Housing Recovery and Reconstruction Platform

NGO - Non Government Organisation

GL - Ground Level

EQ - Earthquake

WC - Water closet

NPR - Nepalise Rupees

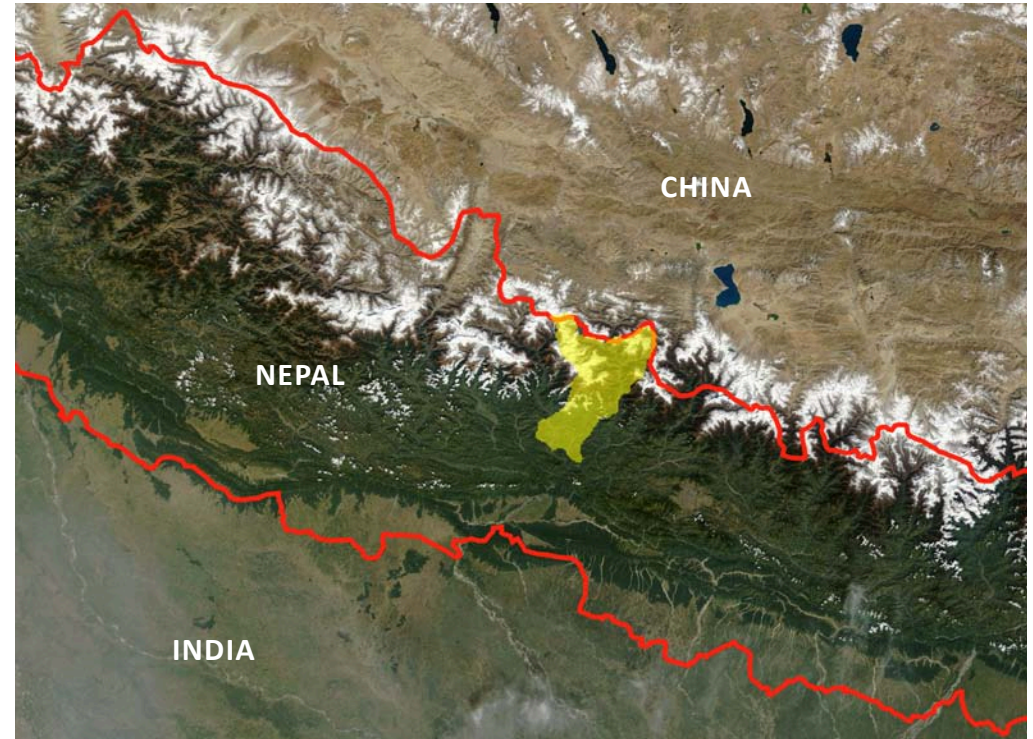
1.1 Introduction

The two earthquakes in Nepal, April 25th and May 12th 2015 respectively, left over half a million homes destroyed and many additional households damaged. The first measuring 7.9 in magnitude struck Nepal between Pokhara and Kathmandu, approximately 80 km northwest of Kathmandu. The worst hit areas are within villages along the mountain ridges within Gorkha. Gorkha district has remote communities which CRS has not assessed for support or programming.

The purpose of the visit was to provide input to the CRS shelter Nepal program by observing and assessing the remote situation in the North. There is diversity in construction types within the 14 working VDCs (a government administrative unit) including RCC, Stone with mud mortar and dry stone masonry. The four VDCs up in the north (Himalaya Region) are not accessible by road, which creates a very different context for support compared to the southern parts of the district. CRS's Nepal recovery and reconstruction strategy has a strong focus on ensuring houses are rebuilt as safer structures and that the affected population has the knowledge and capacity to design and build appropriate, safe and acceptable housing solutions. A range of activities are part of the technical assistance and include skilled and semi-skilled construction training; household level

'Build back safer' awareness training; support projects for locally grown sustainable building materials (such as bamboo); demonstration construction (houses and panels); settlement and village reconstruction planning; CRS has planned to build multiple demonstration houses in each VDC it is working in, including northern Gorkha.

Gaining a good understanding of the built environment, the limitations and opportunities for reconstruction and recovery efforts is essential to not only design an appropriate strategy and formulate advice, but also to inform technical guidance, trainings and demonstration construction etc. to be implemented in the area. Following the principles of good shelter practice the research trip was conducted with a view to involve indigenous materials and vernacular construction techniques. The research trip allowed for rigorous inspection of the construction in the north, to identify some good practices and finalize recommendations to inform programming and designs suitable to the area.



Gorkha District highlighted in yellow

OBJECTIVES

- Assess building typologies and construction techniques, identify good and bad practices, define recommendations for future designs in the area and evaluate techniques for future construction with indigenous material
- Provide recommendation for housing programming in the north for reconstruction, repair and retrofit options, as well as working modalities in the area in terms of field supervision, training and technical assistance.

1.2 Current Support Context

AT TIME OF REPORT

- The National Reconstruction Authority (NRA) was formed in early 2016 and is now the leading authority for reconstruction and recovery in all sectors. It is a new authority with limited access to internal funds and as such is still building its capacity.
- The Department of Urban Development and Building Control (DUDBC) is the lead organisation for the housing sector. However, with the formation of the NRA it is struggling to find its place in the new landscape.
- The Government of Nepal strategic documents, the Post Disaster Needs Assessment (PDNA), the Housing Reconstruction Strategy, and the Post Disaster Recovery Framework (PDRF) have all highlighted the need for support from I/NGOs in the form of technical assistance and guidance to households. Various iterations of the government have meant that the direction from authorities has not always adhered to this guidance. However, the majority of I/NGOs continue to follow this guidance.
- The Government of Nepal (GoN) has committed to provide a total grant of NPR 200,000 (and an additional NPR 50,000 as a 'vulnerable top up') to eligible beneficiaries after they have been enrolled, signed an agreement with the GoN, and opened a bank account. This is intended as a contribution to households for the rebuilding of their house (not the full cost of building). The financial support will be disbursed in three tranches of NPR 50,000, 80,000 and 70,000 based on compliance to the Government structural criteria.
- A total of approximately 200,000 (out of an estimated total of 650,000) houses in the 14 affected districts have already been rebuilt. The structural details most likely won't meet the GoN compliance criteria to allow for instalments to be released. Both I/NGOs and GoN are unprepared for this reality.
- Bank access remains a major impediment to households being able to access the funds from the government subsidy scheme. The majority of people in the north of Gorkha do not have access to banks. Branchless banking will be used in most remote areas, however in areas not accessible by vehicle special security and transport arrangements will have to be made, adding significant cost to the funds dispersal.



1.3 Executive Summary

Each village and Village development committees (VDC) visited has its own unique constraints, for example the availability of materials, the level of community skill-sets, and income. This results in unique details and approaches being implemented. The definition of the home as a single structure has also been challenged, with each household typically owning more than one Khim (Tibetan for house). As the seasons change families move across valleys or to land at different altitudes or orientation, this is tied closely to farming traditions and needs. Approximately 70-80% of houses have been rebuilt with the same details used before their collapse. There was no evidence of seismic improvement details being incorporated anywhere. This lack of compliance to the Nepal Building Code and seismic guidance available is likely to mean the receipt of any government grant will be severely restricted.

This report tries to identify the unique issues present for each VDC and village context and the differences between these contexts need to be carefully considered in any future programming or proposals. The recommendations are split into three distinct groups. The first focuses on shelter retrofit options to make structures safer and increase their resilience to earthquakes. These options are the most applicable as the majority of houses have been rebuilt. The second

includes recommendations that explore details that should be considered when constructing a new house. The third group identifies other recommendations that deal with issues connected to tourism, livelihoods, access, and land management.

It is important to note that the culture of house building is one of relative impermanence and rebuilding sections, or indeed complete houses happens on a regular basis. In this context the idea of permanence refers, roughly, to a time frame of 5 to 30 years. With this in mind, to provide knowledge for and capacity to understand seismic improvement in new buildings could have a positive effect for future safer buildings.

New building recommendations will most likely follow a similar vein to the commonly agreed 10 key messages for safer building in rural Nepal with some small changes specific to the context of the communities.

The structural recommendations for interventions to houses already built generally requires removing the roof to provide a ribbon plate or top plate to stone walls and replacing the heavy stone gable with timber. In areas where timber structures are utilised as a first or second storey it will be important to install bracing. Additional recommendations have also been made,

and are outlined in the report.

It is important to note that no households were reported to be living in temporary shelters, however winter and monsoon preparedness have been identified as issues facing the majority of households, with leaky roofs, draughty walls and other problems. Landslides inhibit access via paths and tracks after monsoon and make import and export of people and goods extremely difficult. These seasonal challenges are expected to affect any reconstruction works, especially the supply of materials from outside the area. It is likely to also hamper the disbursement of the government housing grant and the subsequent compliance inspections for the tranches. Currently the plan by the government is to enrol all northern VDCs in the grant scheme after the monsoon.

Internal air pollution from open fires is another issue noted, impacting indoor air quality and most likely health. Chimneys are usually large voids at the roof level reducing fabric airtightness and reducing internal temperatures and comfort during winter. While this issue is not seen as critical by many households we feel that it is a critical issue to address to ensure safer buildings, and see improvement as likely to make a big difference.

When the road comes! The communities visited all support and want a road

(approx.. 90% of people). Households see it as a crucial step towards access to better services such as hospitals and schools. The hotel owners and people involved in the tourism work realise how a road will be detrimental to their business as the trekking route will be less appealing for tourists. There's talk that it would take 20 years before it could reach the first villages in Prok. But the locals in the tourism sector propose to reopen their grandfather's trail, which was used for trade and access south before the tourism created the valley Manaslu route.

Identified issues and problems have been outlined throughout the report using sketches and photos and in section 7 on page 52 a series of recommendations are proposed for each issue and these will help inform future programme objectives and strategies.

2.1 Gorkha North

OVERVIEW

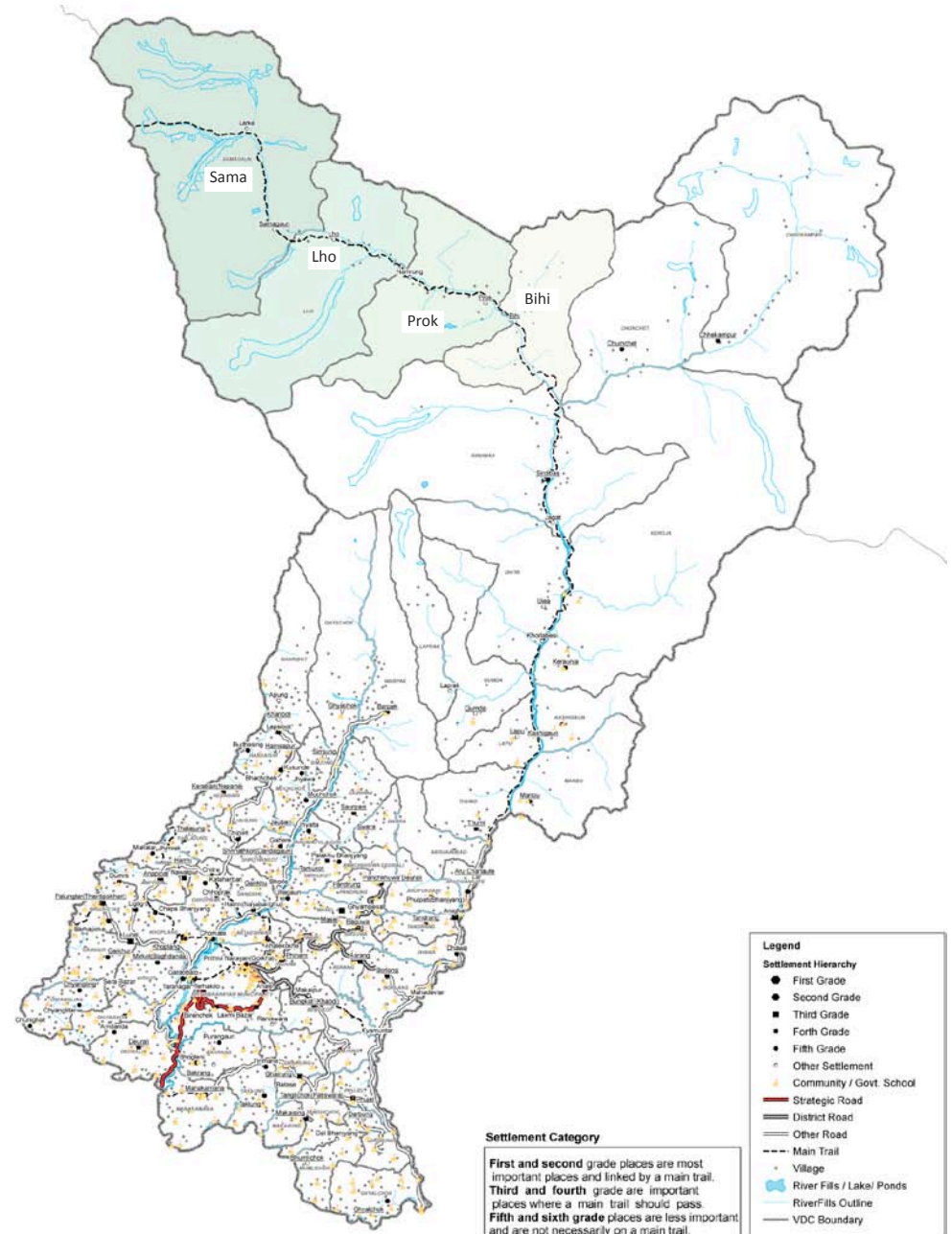
The route from Samagaon to Bihi follows the Mount Manaslu trekking path and is a popular route for tourists. The communities we spoke with say approximately 10% of people in the North rely on tourism and trekking for direct business. But the wider community benefits indirectly as mule owners and porters supplying hotels and as food growers and hotel construction workers. Over the past 5 years the number of new hotels has grown steadily, with every village en-route providing lodging and restaurant services. The hotel industry and the Buddhist education system have promoted the learning of English and Nepali. Tibetan is the first language spoken throughout the valley with some minor dialect differences.

Land ownership is managed within the communities. Very few deeds or documents are kept updated with the government to clearly define ownership or to reflect land being passed through generations. This is likely to be a source of grievance during the government housing grant enrolment process. The population is however very aware of who owns what to the extent that each tree and goat does not need marking to identify its owner. Interestingly, the community has no interest in outsider investment, with no one from outside the community permitted to purchase land. The father traditionally passes

down land and assets to the eldest son and it was common for brothers to share one wife to keep land in the family.

Many, if not most, damaged houses were rebuilt in the first few weeks or months after the earthquakes. This very quickly saw the northern population out of temporary shelter. Three main factors were identified as contributing to this. Firstly, the need to respond to extreme seasonal weather conditions; the monsoon in June-July; and harsh winter November to February. Secondly, a culture of household (male) pride to not show their vulnerability or deficiencies in their previous construction. Often the houses were simply put back together, with some new timbers or roofing materials as needed. Thirdly, with little confidence or reliance on the government in the past the affected population did not expect any additional assistance, instead drawing on their own capacity and resilience.

The majority of Khims in this region have been rebuilt and others were in the process of reconstruction. The affected population could not wait for the subsidy from the Government (if eligible they may receive the 500\$ first instalment with no conditions). If they were eligible then there are 2 lakhs (1=100K) available to rebuild, which would be provided at certain stages of the build, e.g. foundations, walls and roof.



2.1 Gorkha North

OVERVIEW

Most children between the ages of 4 -18 leave the area to attend boarding schools, travelling south of the district and further to Kathmandu. Schools are located in Shyala and Ghap for children up to the age of 10.

The total population of Gorkha district is 67,204. For the four VDC's studied in this report the population is approximately 3-4 thousand people.

There are a number of partner organisations working in the Gorkha north area. They include Nawari Nepal who are providing water systems. The MCA/P (Manaslu Conservation Area Project) who maintain tourism infrastructure, are implementing a stove installation program, dealing with water management and running forestry programs. After the earthquake the priority for MCAP and the trekking companies was to reopen the paths that had been destroyed and in places create new routes. Mountain Child (NGO) is constructing education and health centre facilities and PIN (People in Need, Czech Republic) have constructed a model house exhibiting good earthquake resistant details. The Gorkha Army are working in a number of villages and have constructed a mill plant and are planning a number of micro hydro schemes.

Environment + Biodiversity

There are about 9,000 inhabitants

living in MCA and 2,000 species of plants, 33 mammals, 110 birds, 3 reptiles and 11 butterflies in 11 types of forest have been reported from the area. Climate change affecting the Manaslu area has been recorded in the form of usual winter rains vanishing and being replaced by dry winters for the last couple of decades. Farmers have observed changes such as early flowering and ripening of crops and fruits by an average of two weeks. Southerly winter fogs have extended to Himalayan foothills, and haze is more intense over Nepal's mountain regions. MCAP noted the following actions to deal with climate change:

Given Nepal's geophysical conditions, it can benefit by:

- Promoting water saving technologies, models and practices
- Preventing erosion and soil loss through improved land-use practices promoting research and education on climate change
- Developing alternative road access, such as rope ways
- Expanding forest cover and participating in carbon trading setting policies to promote and transfer efficient renewable energy technologies.¹

1 - <http://www.ntnc.org.np/thematic/climate-change-cc> - Date accessed 160523



3.1 Typical Khim (house) Study

DESCRIPTION + FLOOR PLANS

Typical houses or Khims are two storey, double pitched structures, constructed out of dry-stone with no mortar beds to form a rectangular volume, with gables at each end. Intermediary timber floors are not often tied into the stone walls and so do not provide any benefits as a diaphragm. Foundations are typically around 1' deep of un-stepped dry-stone. Verandahs are added to almost all houses with a timber structure set between stone flank or wing walls. These are structurally separate as the stonework is not keyed to the stonework on the main house. Traditionally an independent timber structure sits inside the stone wings to support the first floor. The roof structure is almost always formed using timber in the round. The under-purlins or rafters bear directly atop the stone walls.

All Khims house livestock at ground floor level and the family at first floor level, which seems to be on average 4-5 people. A small attic is found within the roof space and is used for storage. Cooking, living, eating and sleeping all happens within the first floor open plan space and typically the gable end walls have built in joinery providing shelving and storage. In some cases internal timber lining panels are applied to all walls and this helps improve air tightness to the living space. An open fire and griddle is located to one side of the space and a void within the attic floor

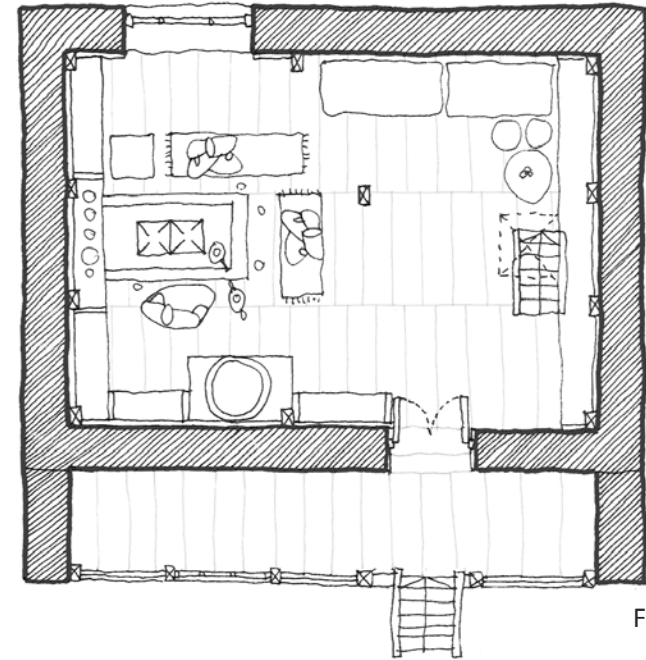
and roof provides an inadequate route for the smoke to egress. A large water bowl contains the drinking water and is traditionally made of copper and located opposite the hearth.

Most villages have a few terraced Khims, on average 5-8 in homes in a row. Some instances adjacent houses share a party wall approximately 18-20" wide and in others two separate walls are constructed. The flank walls to the verandahs are always a party wall construction.

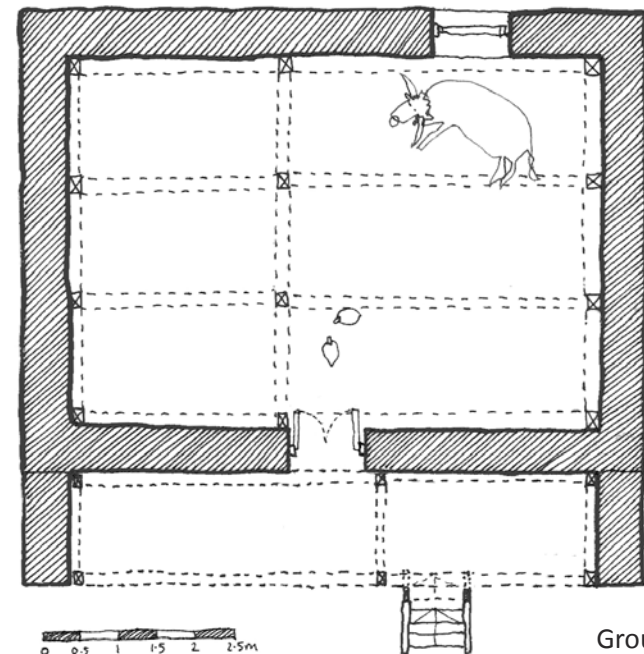
Gutters are carved from a single trunk, with a channel cut along the timber length. No rain water collection was observed. Traditionally roof finishes are either stone tiles approximately 20-30mm thick or timber shingles or planks.

Ornate picture windows are always positioned to provide daylight to the cooking and living space and are typically the most colourful element on the building. Floor is often used to decorate the black timber interiors ceilings and walls with patterns.

Many houses throughout the valley located within villages that were not connected to a electricity supply had solar panels which powered a single light and battery bank and were donated as part of a Chinese support program.



First Floor Plan

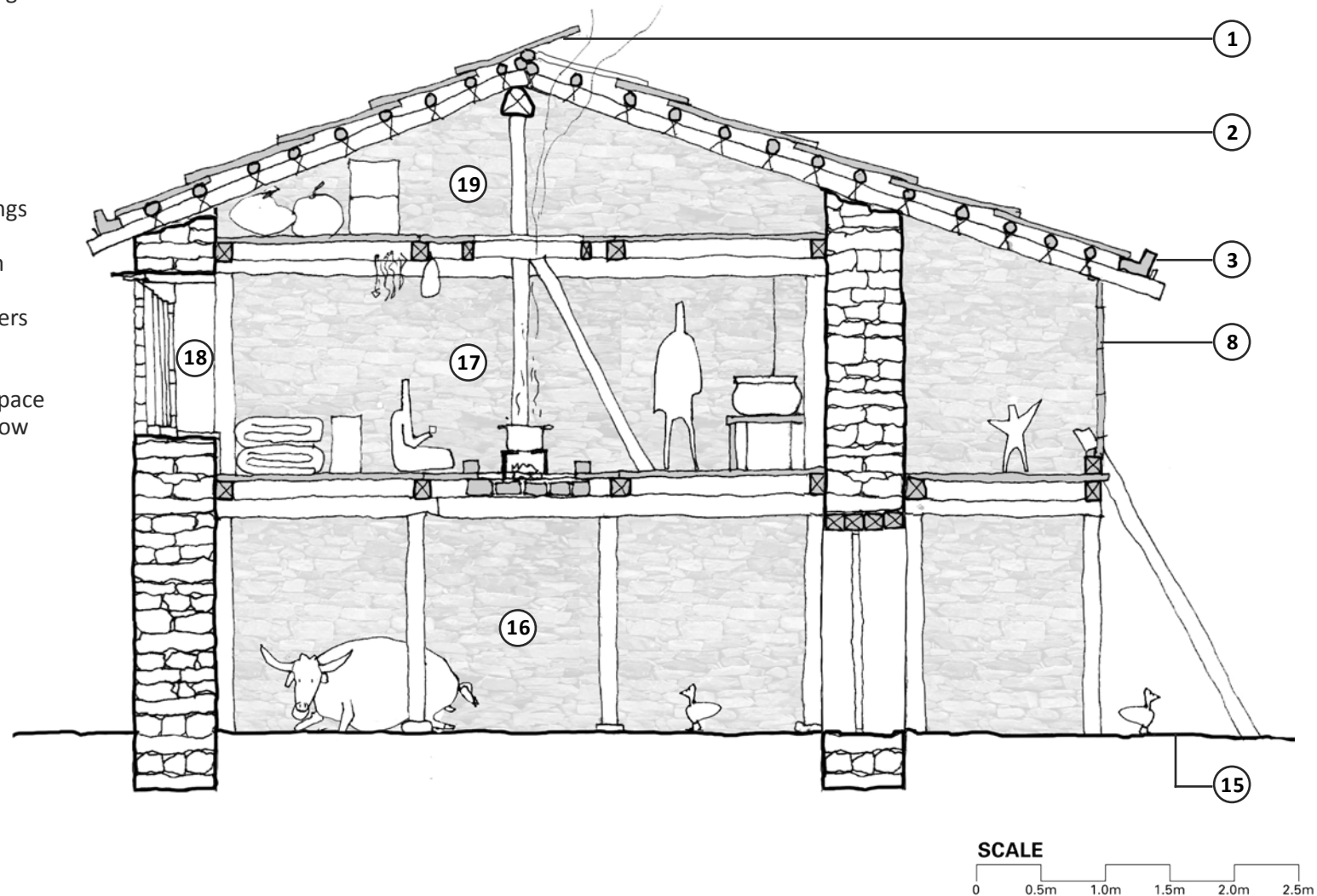


Ground Floor Plan

3.2 Typical Khim (house) Study

SHORT SECTION

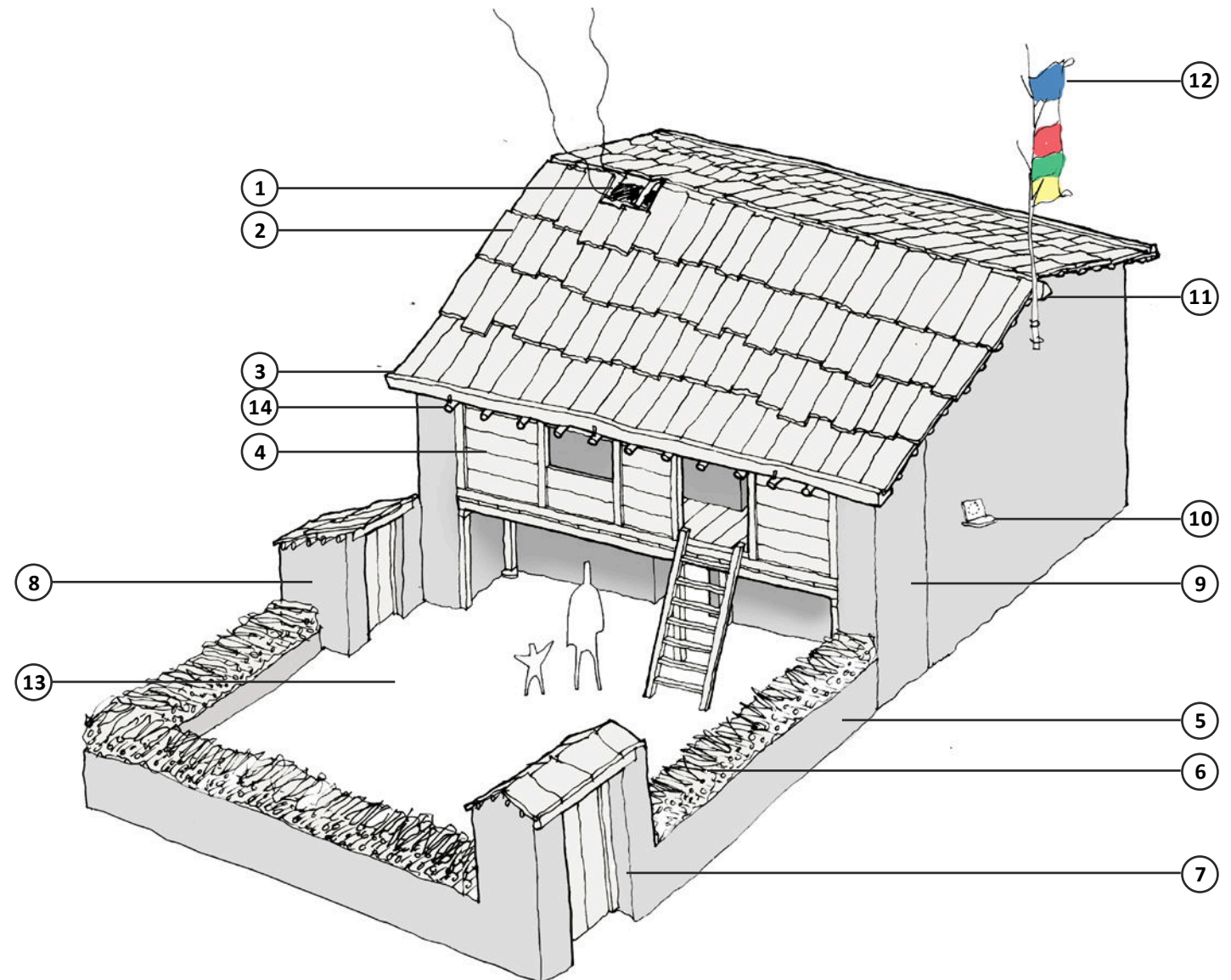
- 1 - Roof void chimney
- 2 - Timber planks or stone tile roof covering
- 3 - Timber gutter carved from single trunk
- 4 - Timber verandah
- 5 - Boundary dry-stone wall
- 6 - Stored fire wood
- 7 - Entrance area
- 8 - Outside WC
- 9 - Separate stone wing wall to verandah
- 10 - Projecting stone shelf for tablet carvings
- 11 - Timber ridge beam
- 12 - Timber flag pole lashed to ridge beam
- 13 - Yard space for livestock
- 14 - Timber pegs fixed to support the gutters
- 15 - Compacted earth floor
- 16 - Ground floor livestock space
- 17 - First floor living / sleeping / cooking space
- 18 - Ornamental carved and painted window
- 19 - Attic for storage



3.3 Typical Khim (house) Study

AXONOMETRIC VIEW

- 1 - Roof void chimney
- 2 - Timber planks or stone tile roof covering
- 3 - Timber gutter carved from single trunk
- 4 - Timber verandah
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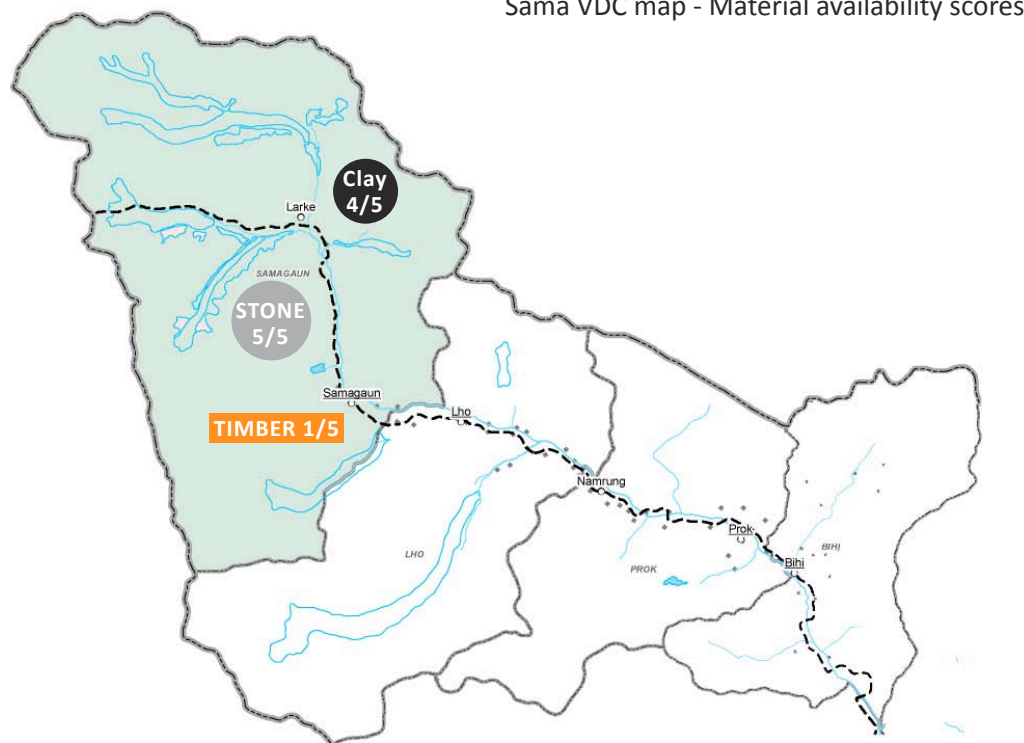


4.1 Sama VDC

VDC CONTEXT

Area	Description	Numbers
	Fully damaged houses (HRRP)	40
	Partially damaged (HRRP)	158
	Minor/ not damaged	2
	Wards	Samagaon 1-6
	Population	Approx. 700
	Households	200 estimate
	Average no. of Khim per household	1
	% Rebuilt	70-80
	Livelihoods	Agriculture / tourism - Hotels / Trading hub with Tibet
Materials	Quality of Stonework	Good
	Availability of Mud/clay	Clay 1/2 hr walk away
	% Use of CGI	20%
	Availability of Timber	Scarce
	Available Timber Species	N/A
	Forest management	Community forest – 1 tree per household, plan to reforest with mix of species
Protection	Winter preparedness	Houses with clay render
	Monsoon preparedness	Households with CGI well prepared
	Water + Sanitation	Water points working
	Temporary shelter situation	None seen
	Health + cooking	Majority of Khims use an open fire indoors

Sama VDC map - Material availability scores



CHALLENGES

- Winter + monsoon preparedness
- Timber expensive and is scarce
- Inadequate timber roof structure to support the weight of stone tiles and snow loads, clear lack of top/wall plates
- High material costs due to remote location
- Stone shaping poor
- Severe cold during winter months, stone walls are draughty
- Inadequate timber used for window lintels



4.1.1 Sama VDC - Samagaon Village

OBSERVATIONS + CONVERSATIONS

Samagaon is the trading hub with China and is on the 'salt to gold' trading route with Tibet. The vernacular style in this village is clearly distinguished with stone tiled roofs. Walls are built using local stone, with some use of clay mortar sourced from ½ hr away. The mortar is applied to the gaps post dry wall construction. Most rebuilt Khims have partly used CGI sheeting (with nail fixings) and this is preferred to stone tiles as it doesn't leak. It was noted that leaky roofs were a major problem during the rainy season. Timber rafters were observed to be under sized in places to support the weighty stone tiles and this is a stability and safety concern.

RCC frame construction with dry-stone infill is used for the majority of Hotels. One hotel was found to have used concrete blocks formed on site and then cement plastered externally. The same hotel used steel poles for the roof structure and wire to secure rafters through the block wall to prevent uplift from high winds.

It was observed that the gable end stonewalls are a common point of failure causing collapse. The size of lintels above windows has in places been observed to be inadequate to take the load of the stonework above and has been noted as a common point of failure. A small amount of stone banding was observed on a lower section of a gable end wall on a single house in the village.

Most people have one house within this community and 50% of the Khims were in a terraced arrangement, oriented with the verandahs facing south.

Timber for construction is scarce and due to arguments with the Lho village community they are unable to trade with them for timber. A conflict arose due to Lho community harvesting caterpillar fungi on Samagaon land. Each person within the community is allocated their own trees and there are plans to plant a community mixed species forest.



Figure ground plan

CONVERSATIONS



VDC warden – Ache Lama - Evening conversation was had with Ache Lama, they were thankful for CRS NFI support, which included thermal mats and CGI during the emergency response. They explained their main concern was leaky stone roofs during the monsoon season. The structure was newly rebuilt with a traditional layout with livestock at ground and living/sleeping/cooking upstairs within a single space. A single solar powered light hung from the ceiling to illuminate the stove space. Attic was used for storage.

4.1.1 Sama VDC - Samagaon Village

DETAILS + TECHNIQUES



Collapsed gable end wall due to lack of through stones



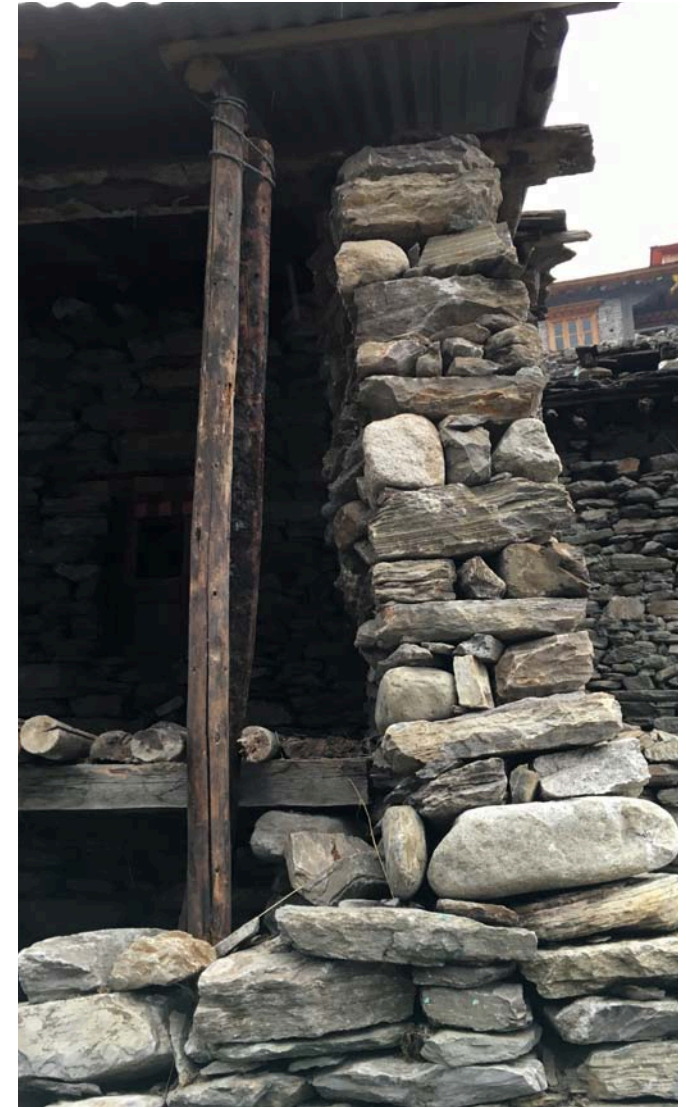
Portion of wall with large, long and well stones survived



Top of wall collapsed around a window opening, lintels are often under sized to take stone wall above



Kuruga Masons employed by households to rebuild and were observed laying through stones



Bad example of a newly reconstructed Verandah flank wall, leaning, stone size and shape inadequate

4.1.1 Sama VDC - Samagaon Village

DETAILS + TECHNIQUES



Clay mortar applied by hand to newly reconstructed Stupa



Complete stone house without a verandah that survived the EQ. Extensive use of clay mortar



Good shape and size of stonework but wall bowing on the left. Traditional stone tile roof with CGI over the Verandah



Run of terraced Khims, with boundary stone walls



Clay applied to gaps between stone work

4.1.1 Sama VDC - Samagaon Village

DETAILS + TECHNIQUES



Stone banding, piece laid approx. 1m above GL and is 100x deep x 2.5m long



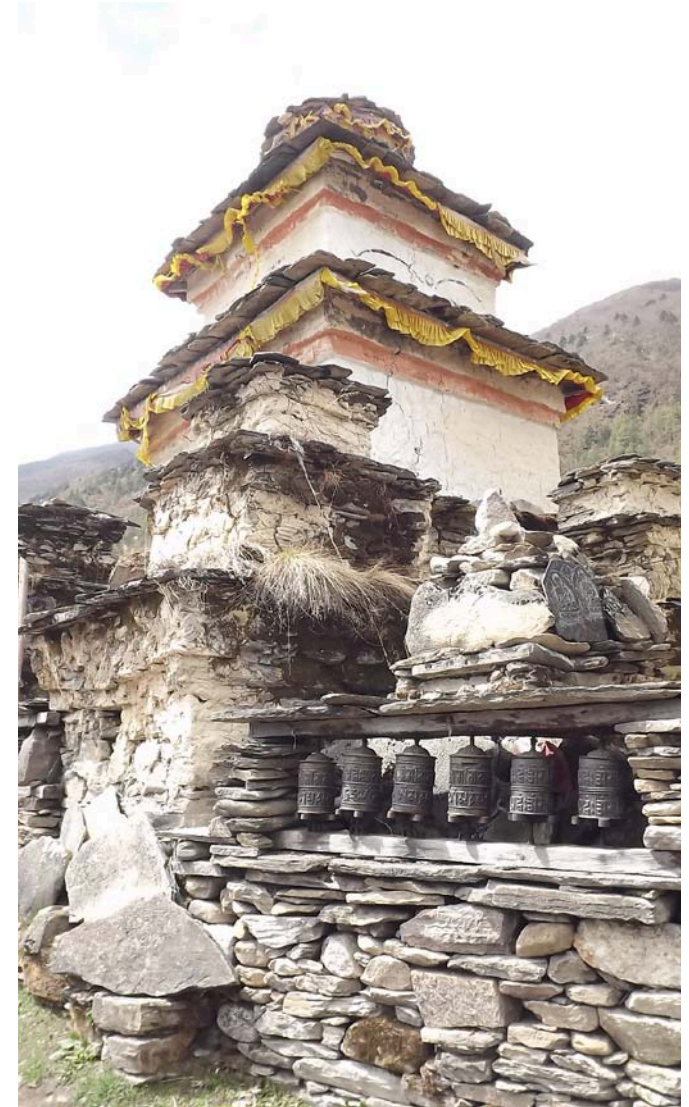
Use of steel pole and hooks on a concrete block hotel, wire wall ties used to secure the rafters



Carpenters working with hand tools



Best examples of stonework were observed on the hotel buildings



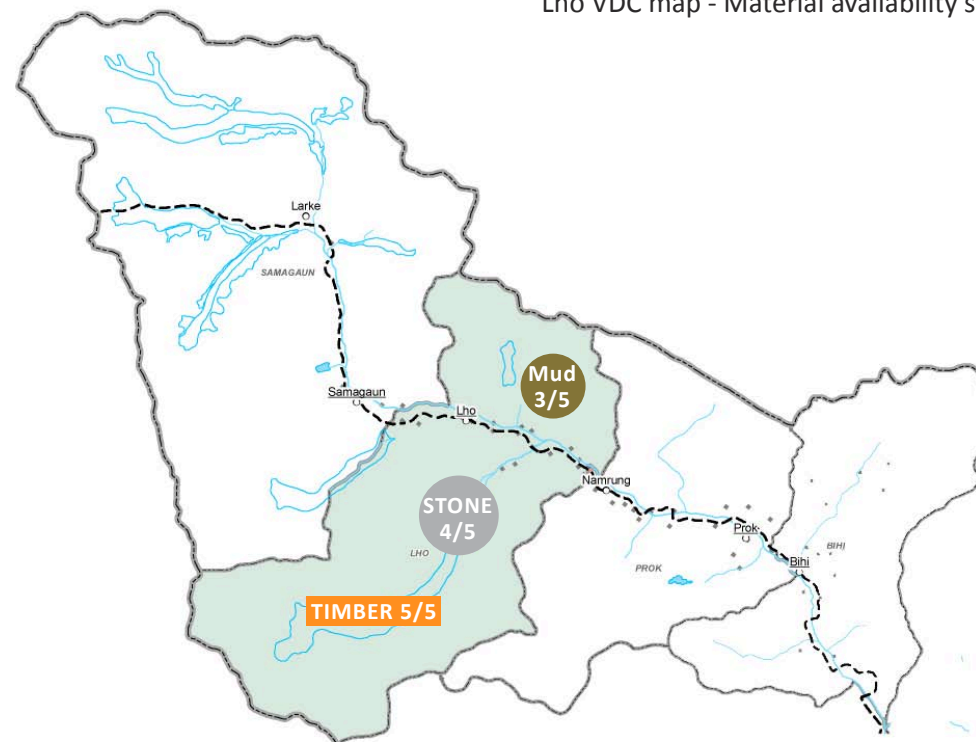
Dry-stone used for all the monument structures throughout the village

4.2 Lho VDC

VDC CONTEXT

Area	Description	Numbers
	Fully damaged houses (HRRP)	282
	Partially damaged (HRRP)	0
	Minor/ not damaged	0
	Wards	Shyala TBC / Lho 6-7-8
	Population	1,000 Lho / 150 Sho / 300 Shyala (approx.)
	Households	300 Lho / 40-50 Sho / 70-80 Shyala estimate
	Average no. of Khim per household	Lho 70% > 3 / Sho 2-3 / Shyala 2 (winter / summer)
	% Rebuilt	70-80
	Livelihoods	Agriculture / tourism
Materials	Quality of Stonework	Poor on Khims, excellent on Hotels
	Availability of Mud/clay	Mud available
	% Use of CGI	30%
	Availability of Timber	Good
	Available Timber Species	Thang / Ji / Sala
	Forest management	Community + Private owned
Protection	Winter preparedness	Mud render provides air tightness
	Monsoon preparedness	Households with CGI well prepared
	Water + Sanitation	Water points working, (Lho - existing source contaminated and there are plans to construct a new pipe 6km from Shyala to Lho)
	Temporary shelter situation	None seen
	Health + cooking	Majority of Khims use an open fire indoors

Lho VDC map - Material availability scores



CHALLENGES

- Winterisation + monsoon preparedness
- Thermal comfort during winter
- Poor stonework observed
- Bracing within timber frames absent



4.2.1 Lho VDC - Shyala Village

OBSERVATIONS + CONVERSATIONS

The vernacular style within this area use timber shingles/ planks as roof tiles. Timber is in abundance and has been used to create first floor external walls where there are verandahs or for the entire first floor structure. Sawn timber is more common and there are many more examples of timber framed extensions and new builds. RCC frame with dry-stone infill is used on the Hotels only, which are 3 storeys in places. Large timber gutters have been used on all dwellings and this detail is more common than in Samagaon. No rainwater collection from these gutters was observed. There is a reduced use of clay mortar infill on the walls.

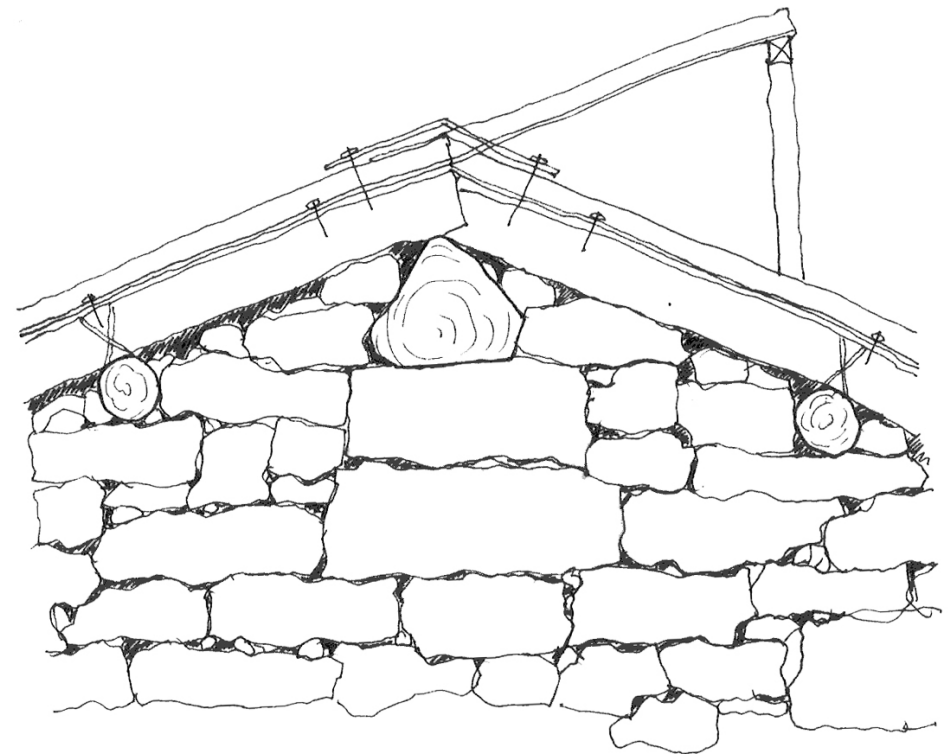
Gable ends and verandah wings were observed as a main point of failure. Some timber banding was observed on the hotels. Surrounding forests are predominately national park, with an application required for tree felling. Different quality of stone shaping/ sizing has been observed with best shaped stones used for the hotel structures. Some houses were seen to have survived and good stonework is most likely a key factor in this.

A large newly reconstructed school sits at the top of the village and houses around 100 students; the schools builder explained his dislike for timber banding, as he fears the rotting wood would weaken the walls.

Large stones have been placed in some instances below the ridge beam, improving stability of the roof structure- see the sketch opposite.



Figure ground plan



LARGE PAD STONES USED TO SUPPORT TRIANGULAR RIDGE BEAM

4.2.1 Lho VDC - Shyala Village

DETAILS + TECHNIQUES



Corner wall collapse with internal timber joinery still standing



Verandah and front wall collapse, roof eave beam supporting the roof on a few stones on the flank wall



Corner wall collapse on a Khim with no original verandah



Verandah flank wall collapse



Stone work to a two storey home verandah wall, bowing outward and tapering in width toward the top

4.2.1 Lho VDC - Shyala Village

DETAILS + TECHNIQUES



Timber post added to support the ridge beam



Timber roof planks as shingles



Large first floor extension in timber



Voids in the roof tiles/ shingles act as chimney's and provide an inadequate escape route for the smoke



Terraces of 3/4/5 Khims are found throughout the village with shared stone party wall

4.2.1 Lho VDC - Shyala Village

DETAILS + TECHNIQUES



Mules transporting cement through the village to supply the hotel building projects



First storey timber found on an hotel structure



Good stonework again found on the hotel buildings, good coursing and stone shape and size



Timber single storey home



Hotel with new first floor timber floor

4.2.1 Lho VDC - Lho Village

OBSERVATIONS + CONVERSATIONS

Common dwellings are built using stone as per Samagaon, but with mud mortar infill on the walls. The vernacular style within this area has timber planks as roof tiles as in the Shayla area. Timber is in abundance and is very cheap.

All new hotel buildings are using CGI for the roof coverings. Lho is the most populous village with over 300 households after Samagaon.

Most households have a small stone outbuilding for their toilet. In some cases a toilet is shared between 2-5 households and this is common along the terraced Khim arrangement.



Figure ground plan

CONVERSATIONS



Fallen trees from the earthquake being sawn on the site of the landslide into four main sizes. Timber planks (floor approx. = 30x300x1500mm), Timber beams (approx. = 50x250x2500mm), timber posts (approx. = 75x75x3000mm). All timber is carried to either Lho, Sho or Lihi.

4.2.1 Lho VDC - Lho Village

DETAILS + TECHNIQUES



Verandah flank wall collapse, these elements are never keyed in to the main building walls



Gable end stone wall and roof collapse



Mud mortar render on a detached small house



Terrace of houses with mud render in the process of being applied



Fully enclosed timber verandah with front yard fenced of for livestock

4.2.1 Lho VDC - Lho Village

DETAILS + TECHNIQUES



Make shift saw mill erected on the site of a landslide



Blue CGI used extensively across rebuilt homes



Timber shingles is the traditional and village vernacular style roof finish



Open chimney and solar panel arrangement on a new CGI roof



Carved timber gutter channel

4.2.1 Lho VDC - Lho Village

DETAILS + TECHNIQUES



Fabric used to lash rafters and battens together



View of the ridge beam and rafters supporting a traditional timber shingle roof, voids in the building fabric can be clearly seen



Chimney void within the roof located adjacent to the gable



Good example of well shaped and sized corner stones used on a hotel building



A single piece of timber banding used on a hotel wall

4.2.2 Lho VDC - Sho Village

OBSERVATIONS + CONVERSATIONS

During the monsoon period people fear landslides and difficult path access. We asked a Sho villager, Tashi Chuden how vulnerable people are identified within the community? She didn't know, but explained that food and board is provided etc. if poor, by the community. Most families have at least two houses as one is for winter and one for summer and many have 3. Tashi explained that she has loaned money to neighbours for reconstruction. Two types of loan are common, firstly where the loan is repaid when she needs help or secondly a loan is given with interest added over a time period (1 lak/ 1 yr= 32k R interest). The monastery has been providing some funds as well as community crowd sourcing initiatives.

The carpenters visited were constructing a large family home with living/cooking downstairs and four partitioned bedrooms above, which is a modern arrangement. The stone walls are around 600-650mm wide and taper towards the roof. A timber frame is situated within the walls to support the first floor and is independent from the stonework. This is described as an 'old style' technique and is the same as the vernacular style in Samagaon. The carpenters bring their own tools and work for payment (4k R per day) and others for a labour exchange; they receive 4 meals and Roksy per day. We were told that the new style keys timber beams into the stone walls.

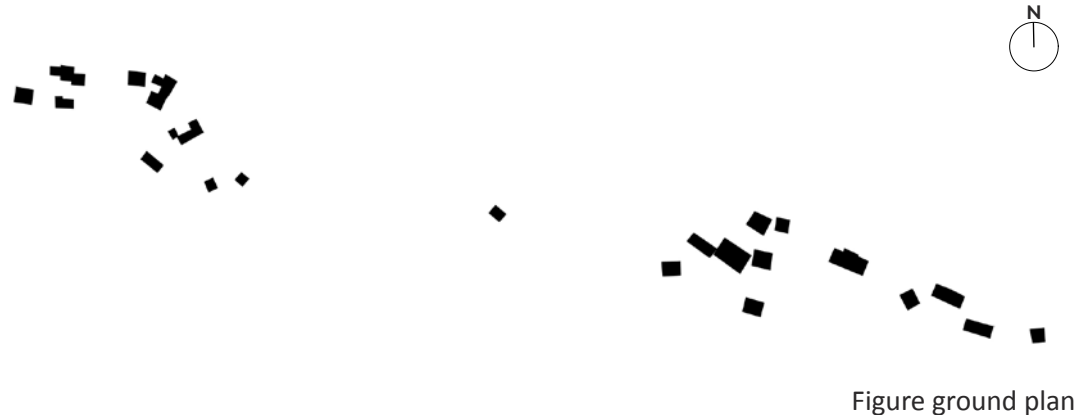


Figure ground plan

CONVERSATIONS



Tashi Chuden - Roof being re-built with CGI, which replaced the timber shingles



House builders constructing a large family home with living/cooking downstairs and four partitioned bedrooms above, which is not the traditional or typical arrangement of spaces.

4.2.2 Lho VDC - Sho Village

DETAILS + TECHNIQUES



Crack observed diagonally in both directions



New house build



Timber being shaped with a hand axe



Arrangement of the internal timber structure of posts, edge beam and secondary floor beams



Flank wall to the house tapers well, but has not been built using a plumb bomb

4.2.2 Lho VDC - Sho Village

DETAILS + TECHNIQUES



Stone shelves project from the wall to support religious scriptures/ mandalas carved into stone



The carpenters tools



Carved timber gutter



A team of locals are laying a new CGI roof natural in colour, the amount of reflection and glare can be seen



Women carrying 3m long sawn timber planks

4.2.2 Lho VDC - Sho Village

DETAILS + TECHNIQUES



View of the timber joinery and timber lined kitchen area interior



View looking towards the single window traditionally located opposite the cooking area



House built into the mountain



View of the attic space used for storage. The gable end wall have large gaps between the stone work letting light through



Stone being collected and transported on the heads

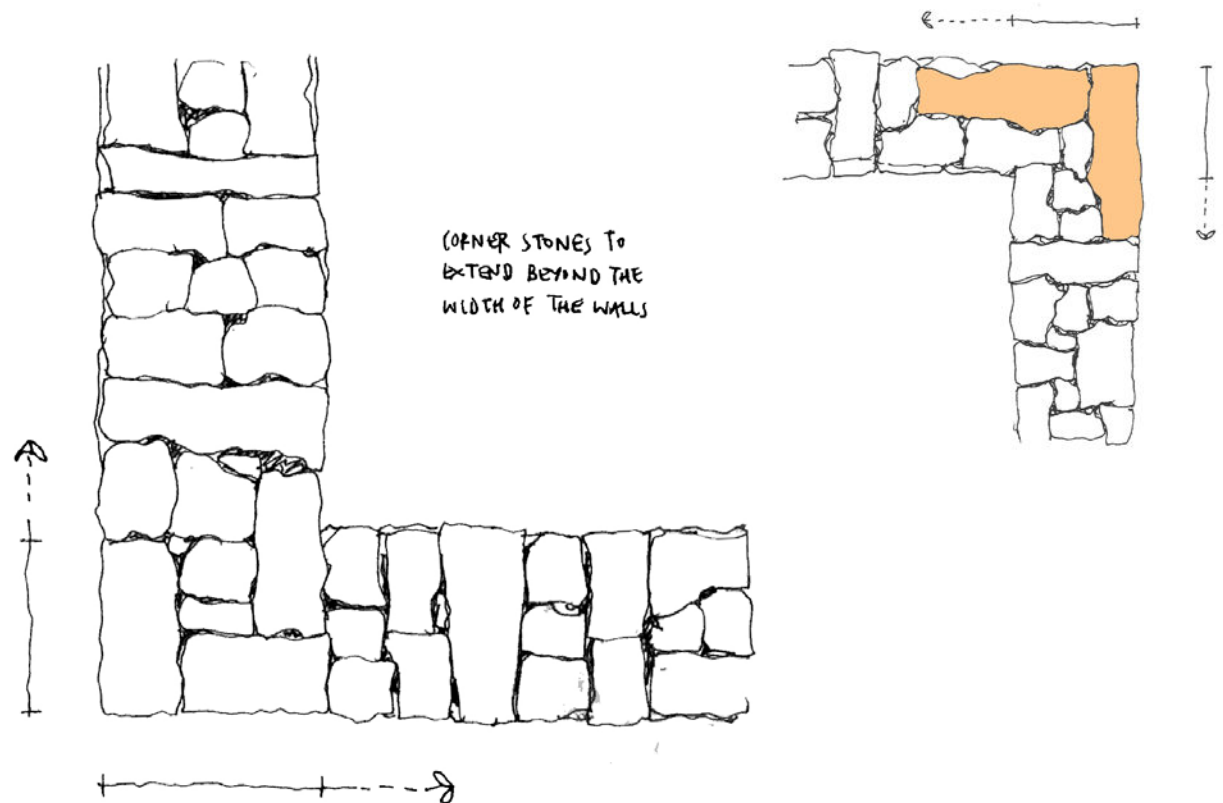
4.2.3 Lho VDC - Lihi Village

OBSERVATIONS + CONVERSATIONS

Lihi is a small village with Khims built in the same style as Lho, Sho and Shyala. Corner stones were observed to be too small on most houses and the sketch opposite shows a good example found on a hotel.



Figure ground plan



4.2.3 Lho VDC - Lihi Village

DETAILS + TECHNIQUES



Verandah flank wall collapse, temporary tarpaulin used for rain protection



Series of carved timber gutters connected to channel water away from the building



Small terraced house with Verandah



Protruding timber first floor element



Large pad stones located beneath ridge beam

4.2.3 Lho VDC - Lihi Village

DETAILS + TECHNIQUES



Timber bands set within the verandah flank wall, clay mortar used at the lower levels



Roof structure supported by a remaining section of the verandah flank wall



Example of poor stonework, round in shape and curving



Existing timber house



Existing timber house

4.2.3 Lho VDC - Lihi Village

DETAILS + TECHNIQUES



Intricate timber details on new temple windows



Cement mortar used to lay new temple walls



Hotel with timber first floor element



Timber retaining wall pins found on newly constructed walls



Plumb bob used to set out stonework on the temple

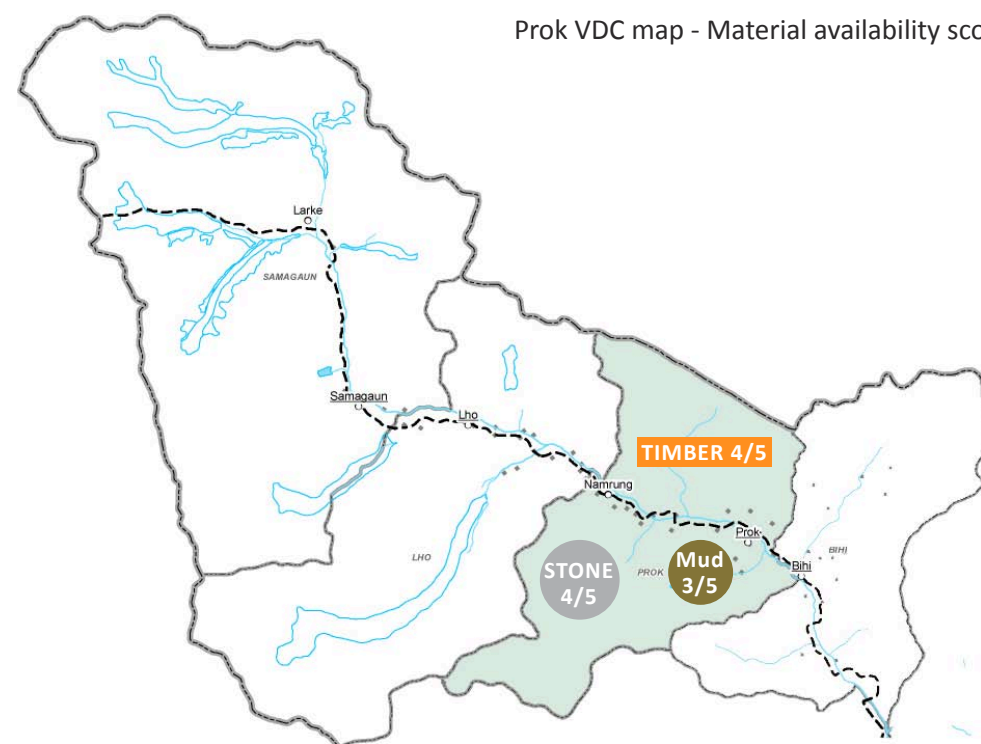
4.3 Prok VDC - Namrung Village

VDC CONTEXT

Area	Description	Numbers
	Fully damaged houses (HRRP)	227
	Partially damaged (HRRP)	0
	Minor/ not damaged	0
	Wards	Namrung 8-9 / Chhak 5-6-7
	Population	150 approx.
	Households	40-50 estimate
	Average no. of Khim per household	2-3
	% Rebuilt	70-80
	Livelihoods	Agriculture / tourism / carpentry
Materials	Quality of Stonework	Poor on Khims, excellent on Hotels
	Availability of Mud/clay	Mud available
	% Use of CGI	30%
	Availability of Timber	Good
	Available Timber Species	Thang / Ji / Sala
	Forest management	Community + Private owned
Protection	Winter preparedness	Mud render provides air tightness
	Monsoon preparedness	Households with CGI well prepared
	Water + Sanitation	Water points working
	Temporary shelter situation	None seen
	Health + cooking	Majority of Khims use an open fire indoors



Figure ground plan



Prok VDC map - Material availability scores

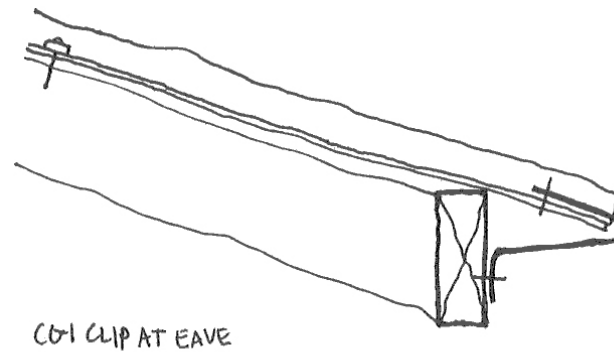
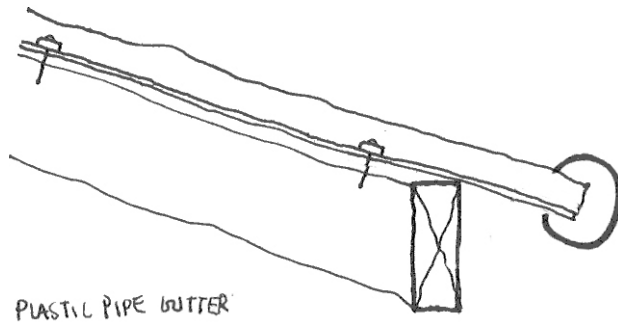
CHALLENGES

- Winter + monsoon preparedness
- Thermal comfort during winter
- Poor stonework observed
- Bracing within timber frames absent

4.3.1 Prok VDC - Namrung Village

DETAILS + TECHNIQUES

Namrung is a small village which had a larger number of hotel/ lodges for tourists and trekkers compared to the other places previously studied along the valley. Below are sketches of two different eave details observed.



Carved timber gutter with steel peg support



CGI strap/clip, see detail opposite



People, stone and timber, the architecture of the Khims reflects the landscape and peoples connection to it.

4.3.2 Prok VDC - Chhak Village

OBSERVATIONS + CONVERSATIONS

This village is spread thinly along the valley and has a small population. The sketch below illustrated the relationship of the traditional timber tiles with its new CGI neighbour.

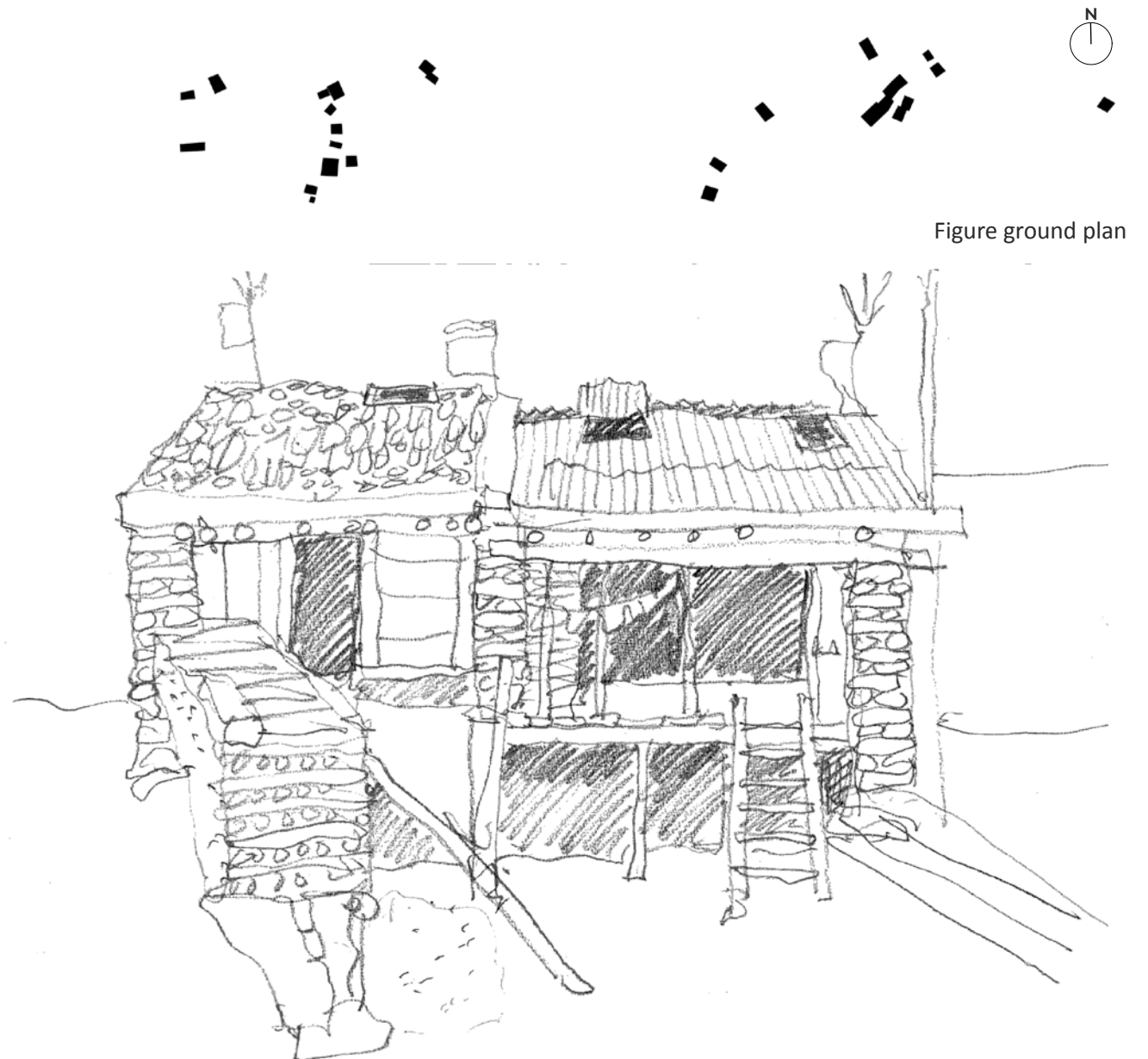


Figure ground plan

4.3.2 Prok VDC - Chhak Village

DETAILS + TECHNIQUES



Collapsed roof structure



Poor stonework, no sign of stones being shaped, bamboo migalo lashing between rafter and purlin



Wall to the left is bulging, carved timber can be seen used for a capping piece over the ridge



Double the quantity of rafters required to take the load of a stone tile roof



Carved timber gutter



4.3.3 Prok VDC - Ghap Village

DETAILS + TECHNIQUES



Good shaping of stone at the lower part of the wall and then very poor at the weakest point within the gable zone



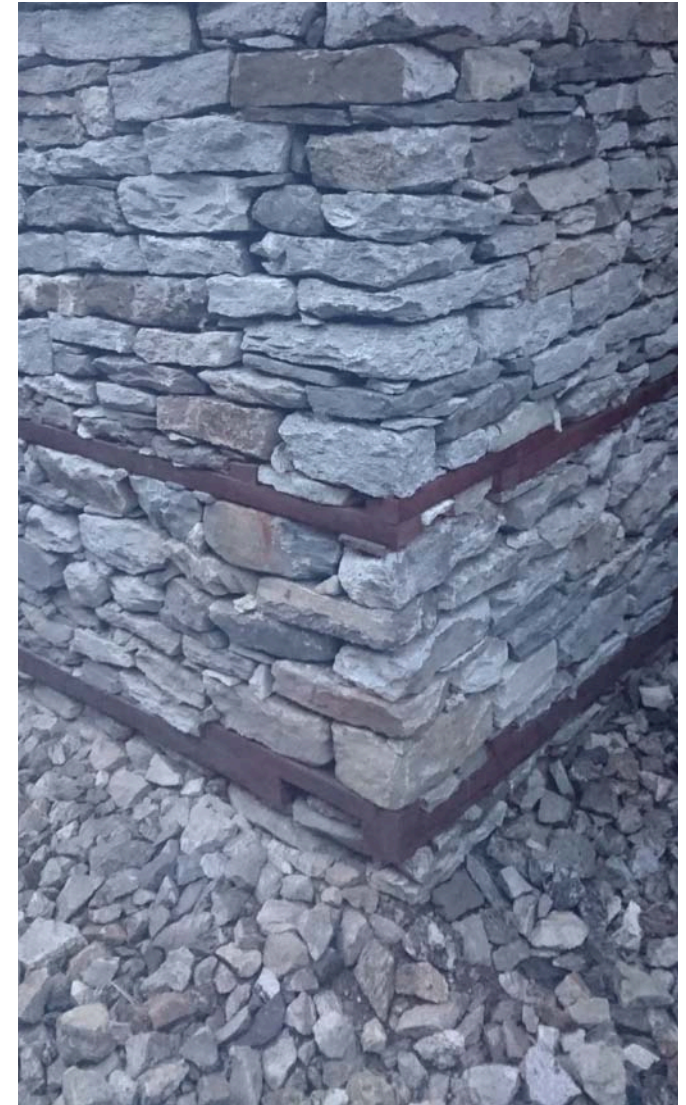
Good shaping and corner long corner stones toward to lower part of the wall



PIN (People in Need Czech R) NGO model house



A creosote paint has been used on all timber elements to help protect from UV, moisture and insects.



Corner stones need to be longer than the width of the wall

4.3.4 Prok VDC - Prok Village

VILLAGE CONTEXT (WARDS 1-2-3)

The village of Prok sits on the brow of a flat slope overlooking the valley, with Khims sparsely spread out, which the figure ground plan opposite illustrates. It was the only settlement visited that had its own electricity supply from the hydro scheme situated to the south.

Many stone gable ends and vernadah flank walls were observed to have failed and the sketch section detail opposite looks to illustrate the typical junction. Wire to secure roof was found on a number of Khims.

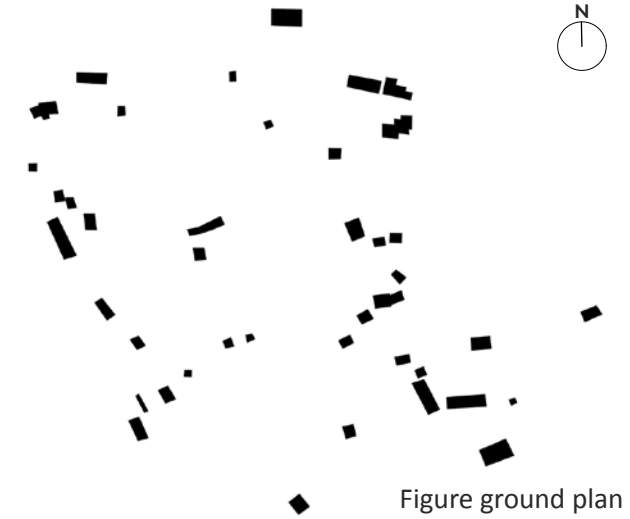
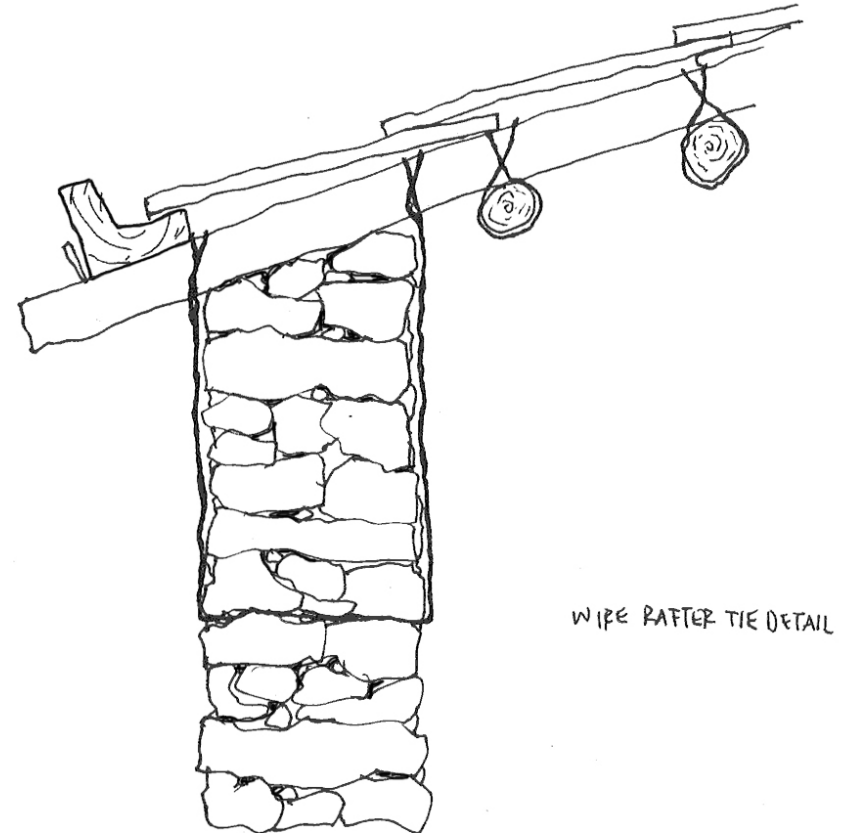


Figure ground plan



WIRE RAFTER TIE DETAIL

4.3.4 Prok VDC - Prok Village

DETAILS + TECHNIQUES



Collapsed corner wall and verandah flank wall, roof temporarily propped with a timber post



Party wall between two properties can be seen on the right with no keystones used



New timber extension to a hotel with no bracing



Temple with stone tiled roof, mud render and adequate quantity of rafters



Timber banding found on the Stupa, an example of the traditional use of this detail

4.4 Bihi VDC

VILLAGE CONTEXT - WARDS 1-2-3

Area	Description	Numbers
	Fully damaged houses (HRRP)	132
	Partially damaged (HRRP)	0
	Minor/ not damaged	0
	Wards	Namrung 8-9 / Chhak 5-6-7
	Population	TBC
	Households	TBC
	Average no. of Khim per household	2-3
	% Rebuilt	70-80
	Livelihoods	Agriculture / tourism / carpentry
Materials	Quality of Stonework	Poor on Khims, excellent on Hotels
	Availability of Mud/clay	Mud available
	% Use of CGI	40%
	Availability of Timber	Good
	Available Timber Species	Thang / Ji / Sala
	Forest management	Community + Private owned
Protection	Winter preparedness	Mud render provides air tightness
	Monsoon preparedness	Households with CGI well prepared
	Water + Sanitation	Water points working
	Temporary shelter situation	None seen
	Health + cooking	Majority of Khims use an open fire indoors, more stoves and flues installed compared to northern villages

Bihi VDC map - Material availability scores



CHALLENGES

- Winter + monsoon preparedness
- Thermal comfort during winter
- Poor stonework observed
- Bracing within timber frames absent

4.4.1 Bihi VDC - Bihi Village

OBSERVATIONS + CONVERSATIONS

Bihi was the last village visited on the trip and was the smallest in size. Stone plinths were more common which formed the base for a timber first storey.

The Gurkha army had constructed a single storey mill which was RC framed with stone and cement infill.

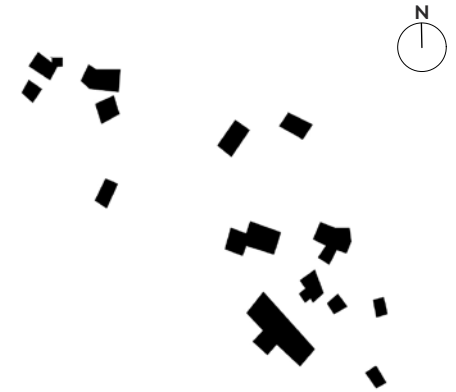


Figure ground plan

CONVERSATIONS



Family of six visited in Bihi, their Khim had a unique layout with the cooking hearth located centrally in the plan



House builders constructing a large family home with living/cooking downstairs and four partitioned bedrooms above, which is not the traditional or typical arrangement of spaces.

4.4.1 Bihi VDC - Bihi Village

DETAILS + TECHNIQUES



Timber post footing with no secure fixing



Chimney void within roof



Hearth cooking area lower than first floor level and is supported by large timber floor beams



Arrangement of timber posts and beams within livestock ground floor area



No fixings between timber elements

4.4.1 Bihi VDC - Bihi Village

DETAILS + TECHNIQUES



String used to tie post



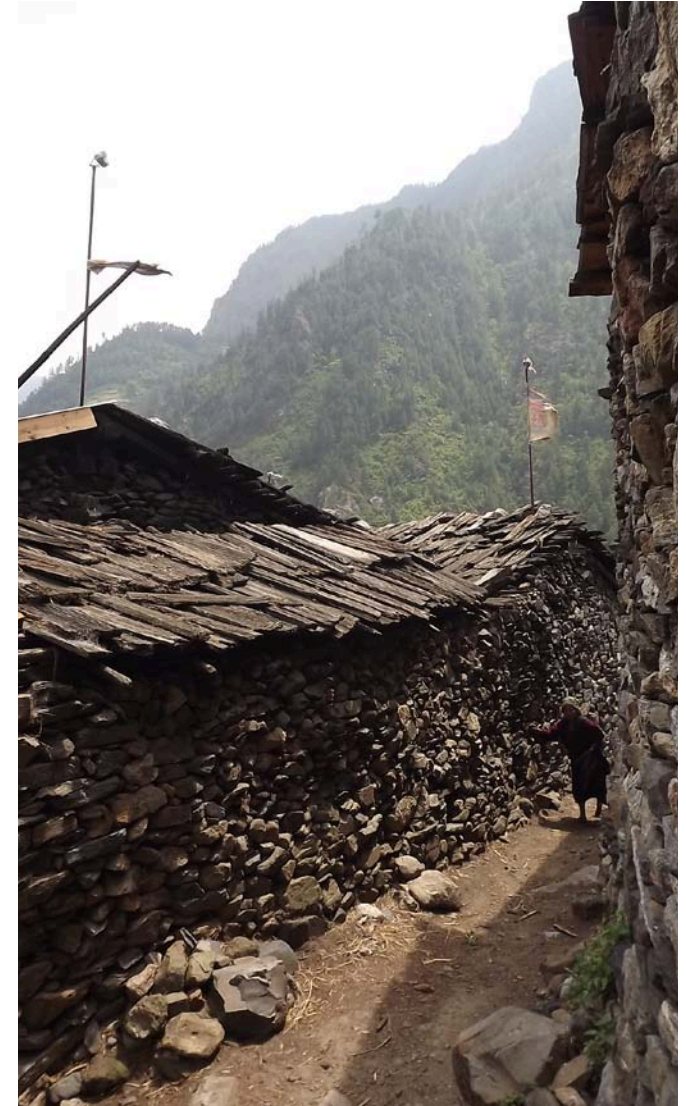
Stone used to weigh down CGI roofing



No fixings between verandah post and ground



New milling plant built by the British Gurkhas



Timber roof shingles and integral part of the village tradition



5.1 Cultural Heritage

CHANGING AESTHETICS

The vernacular construction style of the Gorkha North communities is unique due to their remote location. Traditional skills and construction techniques should be supported in any future programming. Careful consideration of the historical connection between landscape and building is required to protect the heritage of the area. Cultural identity is closely associated to the architecture of the villages, with rituals and traditions informing the use of colour, layout and ornamentation.

The dominant use of stone and timber as building materials strengthens connections between the village and surrounding natural landscape. In essence the buildings are the mountains and forests. Historically no alien

construction materials would have been imported, but this is no longer the case, with CGI replacing stone and timber tiles at an increasing rate. The percentage of homes adopting this alien material has grown drastically since the earthquake. How is this changing aesthetic of the villages affecting the people's connection to their traditional skills and livelihoods? How is this changing aesthetic, going to affect tourism through the shift in aesthetics and changing style?

The photos opposite were taken in 2011 and illustrate how the villages blend into the landscape. In stark contrast, the photo below was taken in 2016 and shows how the use of CGI drastically changes this relationship between building and landscape.



6.1 Building Codes + 10 Key Messages

NEPAL NATIONAL BUILDING CODE - DUDBC

Khim constructions throughout all the villages visited do not meet the DUDBC building codes. The following review of a typical house analyses noncompliance against relevant codes.

NBC 109+202+203+204:1994 - MASONRY : UNREINFORCED

3.1 - Use of boulder stones have been seen in a number of houses.

3.2 – Quarry stones recommended to be no smaller than 50 thick x 150 long in breadth, many houses were observed using smaller stones within the wall build-up.

3.8 – All timber is untreated and unprotected. Softwoods are being used for structural columns and beams .

4.2.2 – Houses and villages are typically located in open areas, but some houses are at risk of landslides. Sho and Ghap had some house positions where this was identified as a hazard.

4.2.3 – Boulder hazards exist in a number of locations.

5.2 – The Khim vernacular style has good proportions, breadth/ length ratio does not exceed 1:3 and the building heights do not exceed 3 times the width, GOOD!

5.3 – Floor to floor heights are below 2m and do not meet the regulations

6.1 – Strip masonry footings foundations do not step and are on average only 300mm deep, on medium soil type 750mm is recommended. Many houses are built directly off a hard rock bed and therefore don't require foundations.

7.1 – Walls not constructed true and have not used a plumb line. Lack of through stones an issue, shaping and size of stones an issue at corners.

8.1 – Size of openings GOOD.

8.2 - Position of openings GOOD, not near corners or junctions.

9.1 – Seasoned and treated timber to be used for structural elements, hardwood species preferred over softwood, softwood used in most examples.

9.2 – Circular posts preferred, once bark and sapwood has been removed, when sawn timber is used it should be square in section. Timber posts should be suitably fixed to a stone base pad. Timber posts in a horizontal line should be founded on a timber no smaller than the post size, this detail was not observed. No post capitals and no fixings observed between posts and beams.

REVIEW OF 10 KEY MESSAGES FOR MUD MORTAR CONSTRUCTION



The following comments review the relevance of the 10 key messages for the visited Gorkha North VDC's.

- 1 - Technical advice is hard to find within the VDC's visited. The PIN model house offered the best place to gain best practice reconstruction advice.
- 2 - No banding observed in re-built houses, use of timber bands not believed to be durable.
- 3 - This is a relevant message and the majority of re-built houses could improve the size of stones.
- 4 - Stone sizing and shaping is the condition to be considered and emphasized.

5 - Very relevant message as common point of failure.

6 - Wall plates non existent on Khims and is a key message to get across for future new house or retrofit reconstructions.

7 - Floors within Khims are rarely tied to walls and this is a detail that would need to be understood and accepted by the community.

8 - Khims are always two storeys, but the tourist lodge designs need to consider shape and volume carefully as they vary greatly in layout and height.

9 - Relevant message

10 - Foundations not observed in great detail but still a relevant message

7.1 Recommendations

GROUP 1 - RETROFIT OPPORTUNITY

	ISSUE	RECOMMENDATION	
1	Roof structure movement creates weaker walls as stones will be forced to move at the top of walls.	<i>Use a top/ wall/ ribbon plate - Timber ladder should be constructed along the top of the dry-stone wall providing opportunity for strong fixings to the roof structure</i>	
2	Gable stonework is unsupported and is a common point of collapse	<i>Use lightweight gables - Timber truss structure should replace the stonework above the top plate</i>	
3	No bracing has been observed to help improve the structures rigidity	<i>Use bracing - Timber structures require bracing in two directions and should follow the rule of thumb illustrated in diagram on p 54</i>	
4	Poor bracing within the roof	<i>Use collar ties between roof rafters and bracing as illustrated on p 54</i>	
5	Ridge beam poorly supported	<i>Use internal posts to support ridge beam</i>	
6	Lintels under sized above window openings	<i>Increase quality and size of timber lintels to support stonework above</i>	
7	Internal air pollution	<i>Replace open fires with stove and flue for cooking and heating</i>	
8	Stone walls at first floor to verandahs collapsing	<i>Replace stonework with timber</i>	

	ISSUE	RECOMMENDATION	
9	Dry stonewalls not airtight and are draughty, increasing heat loss.	<i>Apply mud / clay render to walls to prevent seal voids between stones</i>	
10	Traditional stone tiled and timber shingle roofs leak and are not airtight.	<i>Support the vernacular traditional building style by maintaining stone tile or timber shingles where possible, issues of water ingress could be mitigated by the inclusion of a moisture control membrane (black) beneath. Green CGI is preferred over blue or natural if used</i>	
11	Water is carried a distance from the community water points back to houses.	<i>Rainwater collection should be promoted and storage containers could be supplied and be positioned beneath gutters for each household.</i>	
12	In high winds the roof structure and roof covering are often lifted and damaged by up draught	<i>Rafter wall ties should be used to secure the roof structure to the stonewalls and these can be retrofitted by threading wire through existing dry-stone walls</i>	

NOTES:

The colours down the side try to indicate how costly in time, resources and money certain recommendations are likely to be.

Colour Key:
Low cost
Medium cost
High cost

7.2 Recommendations

GROUP 2 - NEW HOUSE

	ISSUE	RECOMMENDATION	
13	Small and poorly shaped stones reduces wall stability and resilience	Use large stones, which are shaped to rectangular forms. Corner stones should extend beyond the wall width, with through stones every 1m ²	
14	Weak foundations cause instability of entire house	Use stepped foundations at a suitable depth	
15	Many new walls lean are not level and lean in places reducing resistance to earthquakes	Use of plumb bob/ line and horizontal levels to set out building elements during construction	
16	No timber banding observed	Provide details for timber banding, in areas where timber is scarce steel mesh reinforcement bands set between stone courses could be explored as an alternative	
17	Poor stability for floor posts as no ground anchor connection	Timber posts should be suitably fixed to a stone base pad. Timber posts in a horizontal line should be founded on a timber no smaller than the post size, this detail was not observed. No fixings observed between posts and beams	

GROUP 3 - OTHER

	ISSUE	RECOMMENDATION
18	Deforestation, and growing effects from climate change; increased snow melt affecting water table and cleanliness of the water	Support and promote timber use and support forestry programs, community managed and processed
19	Lack of local community seismic construction skills	Support of training programs in carpentry and masonry construction
20	Communities import excessive food items for hotel menus, which are costly and leave a lot of waste packaging	Support the design of menu's that use local produce and recipes
21	Availability of washing facilities	Provide bathing and washing facilities and hygiene information
22	Low availability of fruit	Support fruit tree planting to provide increased opportunity for livelihoods and food security
23	Poor access to information regarding GoN grant and support	To provide access to information services

NOTES:

The colours down the side try to indicate how costly in time, resources and money certain recommendations are likely to be.

Colour Key:

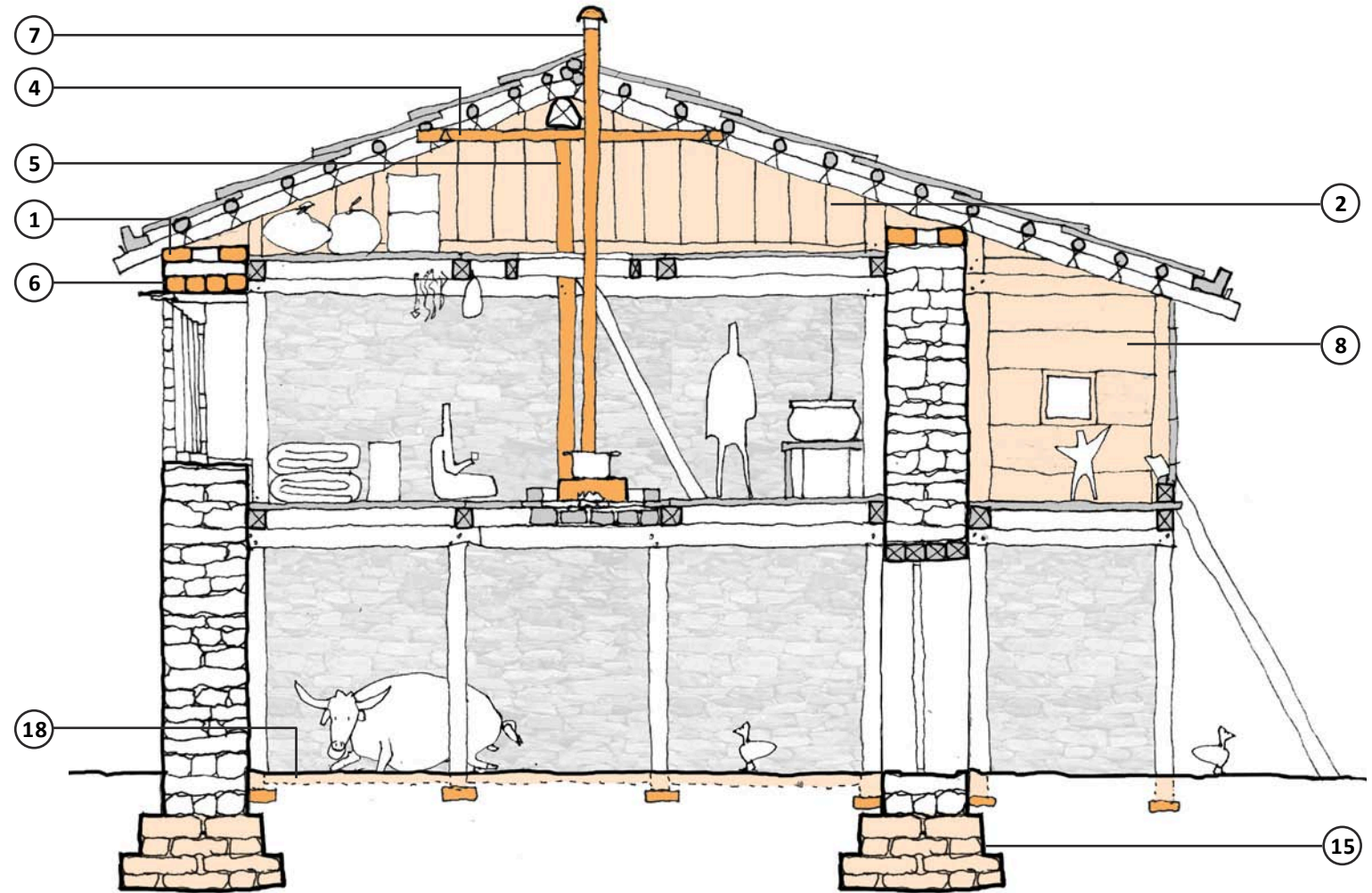
Low cost
Medium cost
High cost

7.3 Recommendations

REVISED SECTION

See recommendations listed on the previous pages.

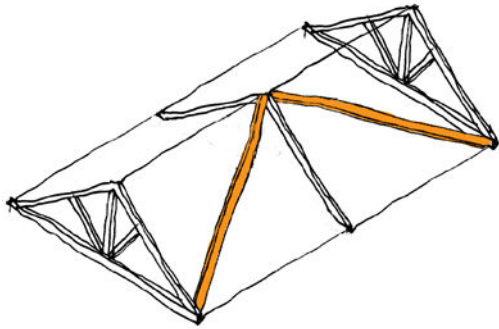
COMMENTS: Some of these recommendations would require retrofitting of the majority of damaged houses which have already been reconstructed. There is no guidance available on how best to execute a retrofitting program. This is partly due to every house or case being unique. The production of approved retrofitting details, through a working group, would be helpful to help assess and inspect construction.



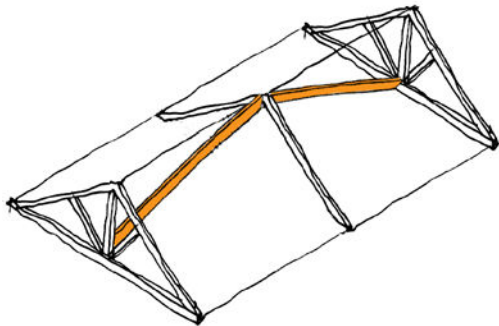
8.1 Details

TIMBER BRACING - ROOFS

Cross bracing can be added to the roof pitch within the plane of the roof rafters

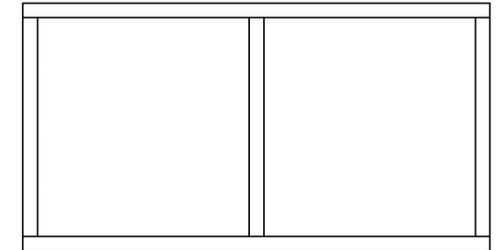


Cross bracing can be added to the roof along the line of the roof ridge.

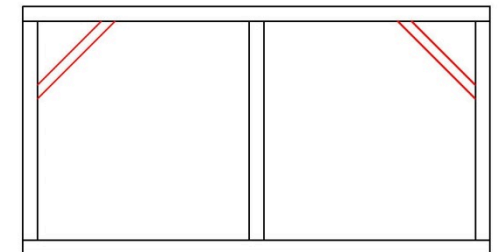


TIMBER BRACING - WALLS

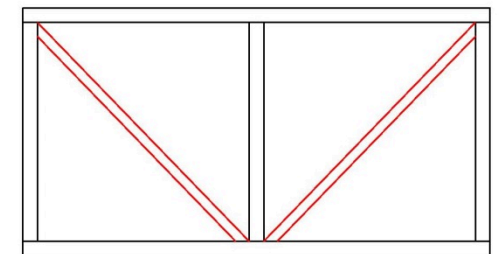
Sheathing or bracing was not observed anywhere in Gorkha North. The diagram opposite shows the simple method of construction adopted with timber studs solely fixed at the sole and head plate.



Knee bracing is OK and is certainly better than none at all.



Bracing top to bottom is by far the strongest and is recommended for new build and for retrofit situations.



8.2 Details

SIZE + LOCATION OF OPENINGS

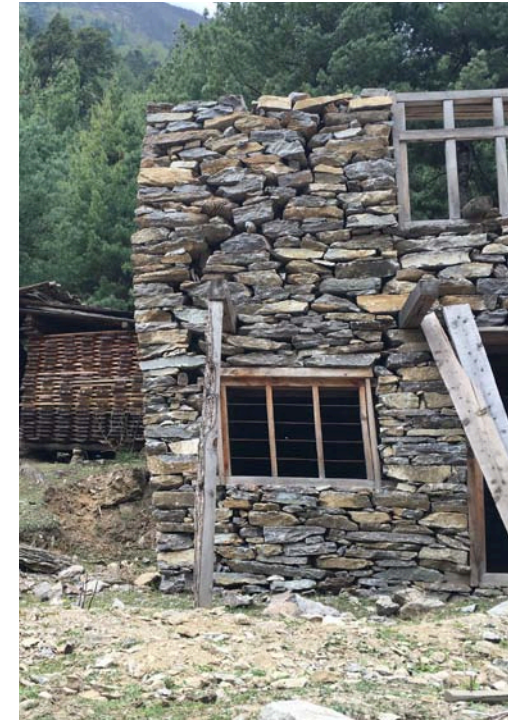
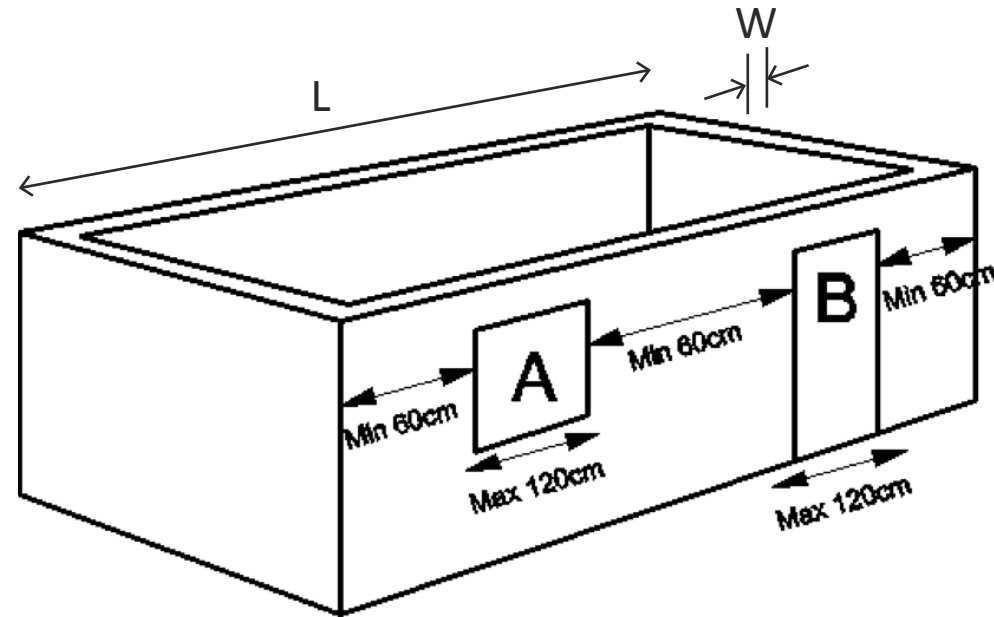
Construction dimensions and details for non-reinforced masonry wall for single storey dwellings.

1. The length of any unsupported wall should not be more than 10 times the width of the wall.
 $L \leq W \times 10$
2. The distance of openings from corners and other openings should be at least 120cm.

3. Door and Window openings should not be greater than 120cm.

4. The total area of wall openings should not exceed 50% of the total wall area.

$$A + B < 50\% \text{ of } C \text{ (total area of wall)}$$



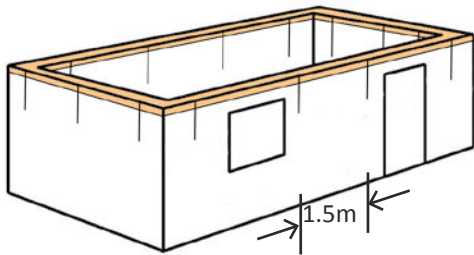
Failing wall with diagonal cracks

8.3 Details

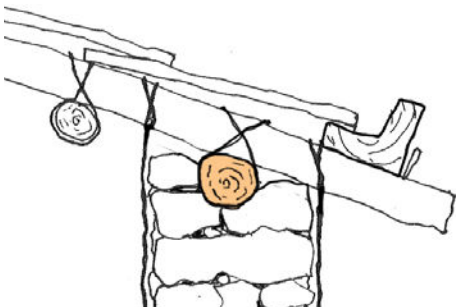
TOP / RIBBON / WALL PLATE

A wall plate can be fitted. If part of the walls collapse, this will support the roof until the wall can be repaired again.

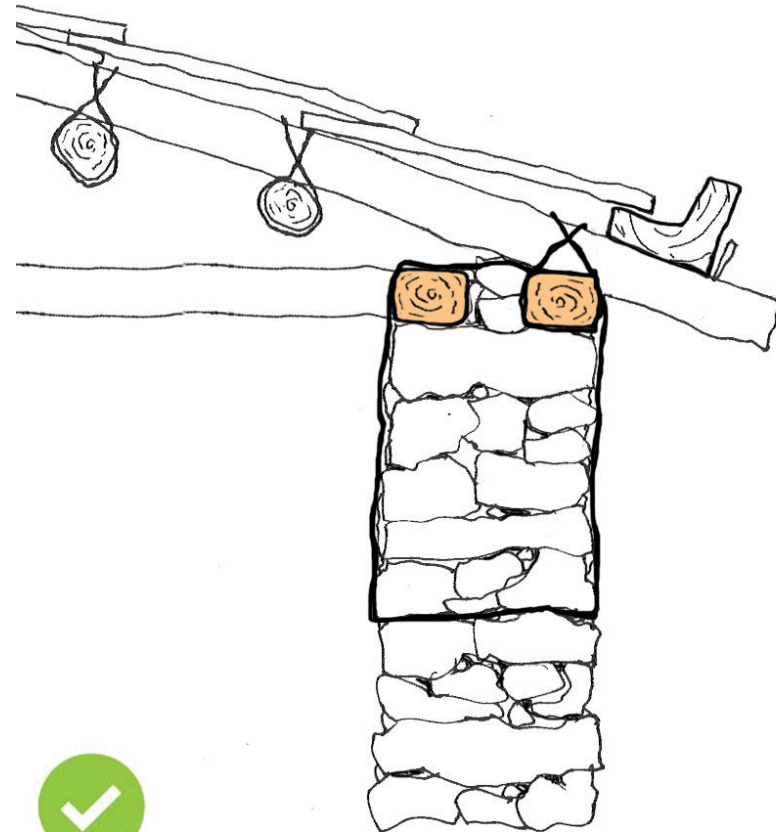
The wall plate should be fixed to the wall with tie wire as shown



Continuous top plate tied through stonework below at approximately every 1.5m.



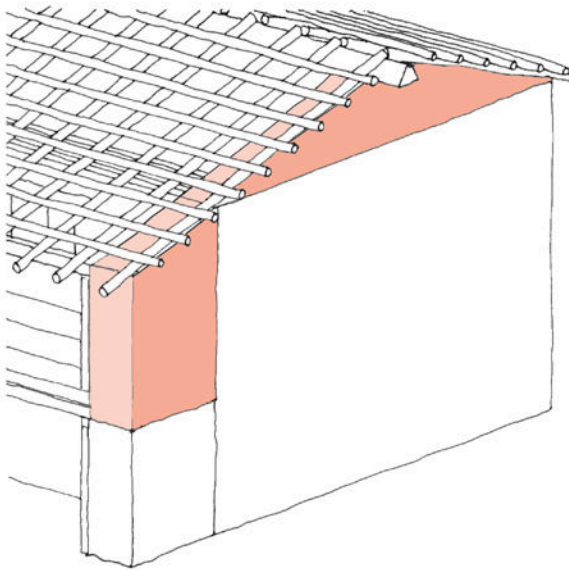
Timber in the round could be used, but not as effective as a sawn timber ladder.



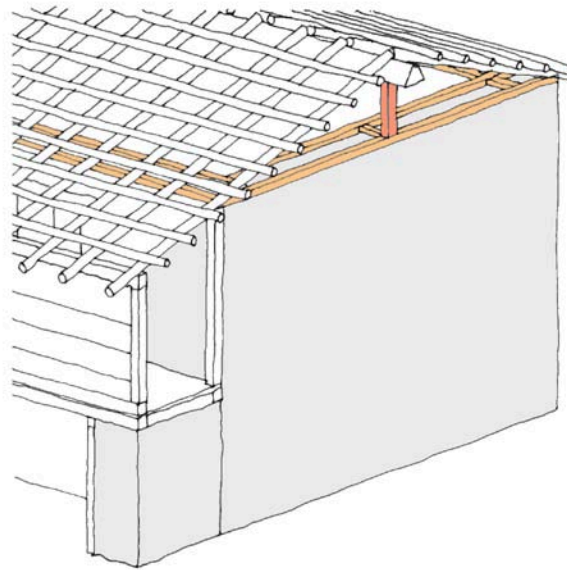
Wall plate formed with sawn timbers connected as a ladder and tied down to the dry stone wall beneath. Roof rafter tied down to outside timber. Timber members should be approx. 75 (h) x 100 (d) mm in size. This is a recommended detail for retrofit improvements.

8.4 Details

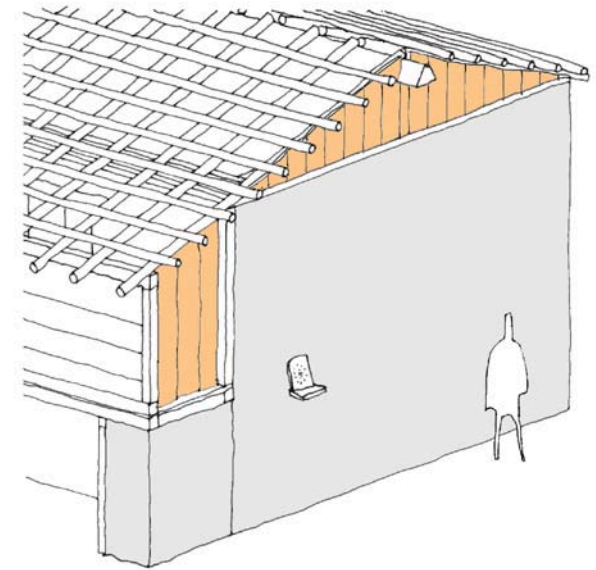
REPLACEMENT TIMBER FLANK WALL + GABLE



Removal of dry stone wall at first floor for both flank walls of the verandah and at the gables.



Addition of a timber top/ wall/ ribbon plate along the perimeter of the wall. A timber post is positioned beneath the ridge beam and bears on the outer timber top plate. Top plate timber sizes should be approx. 75 (h)x 100 (d) mm as illustrated on the section detail opposite. The ridge beam support post (highlighted in red), should be no smaller than a 100mm sq timber, these gable posts are in addition to internal ridge beam support posts.



Timber planks used for cladding as per the details found on the front facade of the verandahs.

9.1 Bibliography

Gabion Bands, Proposed Technology for Reconstructing Rural Rubble Stone Houses after the 2105 Earthquake,
RANDOLPH LANGENBECH, August 27, 2015

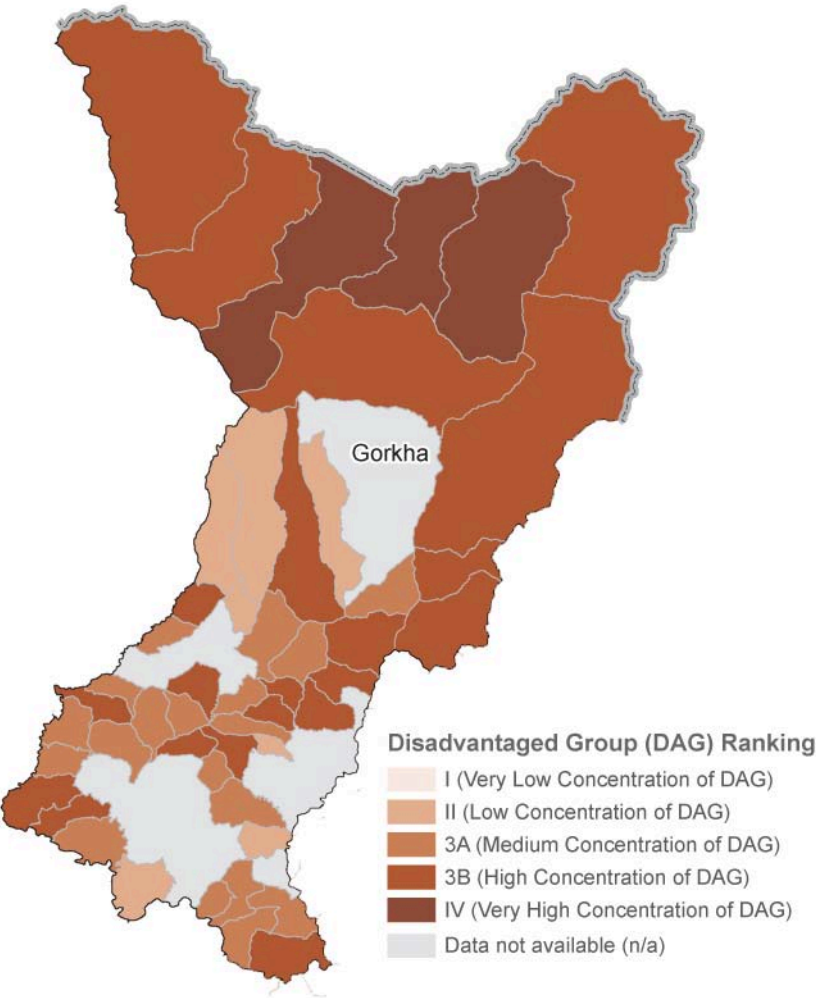
The Nepal National Building Codes 202+203+204 for Bearing-Wall Masonry, Low Strength Masonry and for Earthen
Construction all feature bands either of timber, bamboo or steel for seismic resistance.
DUDBC , 1994

A Tutorial: Improving The Seismic Performance of Stone Masonry Buildings,
JITENDRA BOTHARA + SVETLANA BRZEV, First edition, July 2011

United Nations Educational, Scientific and Cultural Organization, Traditional Materials and Construction Technologies
used in the Kathmandu Valley,
CATERINA BONAPACE + VALERIO SESTINI, Nov 2003

Appendix

I - HRRP - Concentration of Disadvantaged Groups



Appendix

II - Material and Labour Costs Matrix

		Samagaon VDC				Lho VDC								Prok VDC						Bihi VDC					
Material	Type	Samdo (not visited)		Samagaon		Shyala		Lho		Sho		Lihi		Namarung		Chak		Gap		Prok		Bihi		NOTES	
		Rupees	USD	Rupees	USD	Rupees	USD	Rupees	USD	Rupees	USD	Rupees	USD	Rupees	USD	Rupees	USD	Rupees	USD	Rupees	USD				
Stone Shaped	Granite																								
	Basalt																								
	Limestone																								
Stone Unshaped	Granite 1m3		7000		7000		5500		5500		5500		5500		4000		4000		4000		4000		4000	all stone cost same price ,rate varies according to distance 1m h,1m l, 1mw ,	
	Basalt 1m3	N/A					5500		5500		5500		5500		4000		4000		4000		4000		4000		
	Limestone1m3						5500		5500		5500		5500		4000		4000		4000		4000		4000		
Timber	Ji-sing					32-35per ft		32-35per ft		32-35per ft		32-35per ft		32-35per ft		30 -40 per ft		30 -40 per ft		30 -40 per ft		30 -40 per ft		30 -40 per ft	extra transportain that depends upon the distance rate variees as distance of the forest . other species are not used much for housing sama and samdo used this plant but done in daily wages
	Than-sing					28 -30per ft		28 -30per ft		28 -30per ft		28 -30per ft		28 -30per ft		30 -40 per ft		30 -40 per ft		30 -40 per ft		30 -40 per ft		30 -40 per ft	
	Kal-sing					28 -30per ft		28 -30per ft		28 -30per ft		28 -30per ft		28 -30per ft		30 -40 per ft		30 -40 per ft		30 -40 per ft		30 -40 per ft		30 -40 per ft	
	Kil-sing																								
	Takpa																								
	Sukpa (Cedar)																								
	Harsyang																								
Mortar	Clay			Yes																					
	Cement		6700		6700		5200		5200		5200		4700		4700		4450		4450		4450		4200		
	Mud																								
Sand	Local	50-100 per tin		50 -150per tin		50 -150per tin		50 -150per tin		50 -150per tin		50 -150per tin		50 -150per tin		50 -150per tin		50 -150per tin		50 -150per tin		50 -150per tin		50 -150per tin	,mesured mostly in tin 1tin = 20 kg of sand
	Imported																								
CGI	Blue	10200 (46mm) Arugaht rate		10200 (46mm) Arugaht rate		10200 (46mm) Arugaht rate		10200 (46mm) Arugaht rate		10200 (46mm) Arugaht rate		10200 (46mm) Arugaht rate		10200 (46mm) Arugaht rate		10200 (46mm) Arugaht rate		10200 (46mm) Arugaht rate		10200 (46mm) Arugaht rate		10200 (46mm) Arugaht rate		10200 (46mm) Arugaht rate	extra traportation by helicopter fromsalyantar 15perkg by potter sama 200 per ft, lho 180per ft, prok 170 perft, bhi 160 per ft , the rate varries according to sesasons.
	Natural (white)	6800(46mm)Arugaht rate		6800(46mm)Arugaht rate		6800(46mm)Arugaht rate		6800(46mm)Arugaht rate		6800(46mm)Arugaht rate		6800(46mm)Arugaht rate		6800(46mm)Arugaht rate		6800(46mm)Arugaht rate		6800(46mm)Arugaht rate		6800(46mm)Arugaht rate		6800(46mm)Arugaht rate		6800(46mm)Arugaht rate	
	Green																								
Steel Rebar	9mm D TBC																								
	12mm D TBC																								
Cement	50kg bag													6000		6000		6000		6000		6000		6000	
Steel Pole	50mm D																								
Migalo Bamboo	Bundle TBC																								Used for a single hotel roof in Samagaon
Average number of houses (Khims) per household		1		1		2 (winter/summer)		70% have 3						2		1		1 TBC		1		1			

Appendix

iii - ASPIRE Environmental Assessment

